

Savitribai Phule Pune University
Department of Statistics

Fan Chart and its Application in Inflation Management for India and other BRICS Nations



Pushkar Kale (1825)
Mrunmayi Naik (1836)
Project Advisor
Prof. T.V.Ramanathan

A project report submitted for M.Sc. Statistics

September 2020

Certificate

Name:

Signature:

Date:

Acknowledgement

This work would not have been possible without the support and help of many individuals. We would like to extend a big thanks to all of them. We are highly indebted to Prof. T.V.Ramanathan, Head, Dept. of Statistics, Savitribai Phule Pune University, Pune for his guidance and consistent supervision as well as for giving the problem statement and encouraging us for this work. We would also grateful to Prof. U.V. Naik-Nimbalkar for working as an examiner for our internal presentation and for giving valuable feedback. We would also like to express gratitude to all the Teachers and Research Students of Department of Statistics, SPPU for their support and kind co-operation.

Our appreciations also go to our all friends who have willingly helped us out with their abilities. We are eternally grateful to our parents and family for their strong backing and promising support. Words are not enough to describe our gratitude to them.

Abstract

The last two decades have seen a steady rise in the use of Fan charts to study inflation forecasting by various central banks around the world. The Fan chart helps in communicating the future uncertainties in a simple and easily understandable manner to the public. The decision makers can use these charts for planning future policies. In this paper, we propose Generalised Hyperbolic distribution for the construction of a Fan Chart. Traditionally, Fan charts are developed using Split normal distribution. But Split normal distribution does not allow flexibility in parameters other than for the skewness. Generalised Hyperbolic distribution is suggested as it is more flexible and has variable ranges for the parameters kurtosis, shape, etc. The uncertainty bands of the Fan charts for both the distributions are compared on the basis of their width. The exercise has been carried out for all the BRICS countries. The confidence bands for the Fan charts constructed using Generalised Hyperbolic distribution are narrower than that of Fan charts constructed using Split normal distribution. This indicates less uncertainty, a case Fan chart constructed using Generalised Hyperbolic distribution is more efficient. Perhaps, this is the first attempt of using an asymmetric distribution other than the Split normal for the construction of a Fan chart. This study motivates the exploration of other asymmetric statistical distributions in the construction of Fan charts in various domains other than the Split normal.

Contents

1	Introduction	1
1.1	Aims and Objectives	1
1.2	What is a Fan Chart and Why it is useful?	1
1.3	India and other BRICS nations	3
1.4	Overview of the Report	3
2	Literature Review	4
3	Theoretical Discussion	6
3.1	Split Normal Density	6
3.2	Generalised Hyperbolic Distribution	9
4	Data Analysis	13
4.1	Data Description	13
4.2	ARIMA	15
4.2.1	Residual Diagnostics	16
4.3	Vector Auto-Regressive Model	17
4.3.1	Residual Diagnostics	18
4.3.2	Test for Cointegration	19
4.4	SETAR Model	19
4.4.1	Residual Diagnostics	20
4.5	Model Selection	21
4.6	Construction of Fan Charts	23
4.6.1	India : Observed Data	23
4.6.2	India: Future Inflation	26
4.6.3	Other BRICS Nations	30
5	Conclusions	43
5.1	Findings	43
5.2	Limitations and Future	44
	References	45

Chapter 1

Introduction

1.1 Aims and Objectives

1. To study the Indian inflation by using appropriate Time Series Model
2. To implement Fan chart methodology using Split Normal density for the Indian inflation
3. To explore the possibilities of other asymmetric distributions and compare the computation of Fan Charts by using different distributions
4. To study and discuss the theory and implementation of Bivariate Fan charts
5. To study the inflation behaviour of BRICS countries in view of COVID-19 pandemic

1.2 What is a Fan Chart and Why it is useful?

Future direction of the economy is an essential input to any policy making process. The future risks to the forecast are unavoidable due to the various interaction effects between the variables and due to the sudden shocks in the economy. This risk can be quantified by using Forecast interval, which indicates the risk around the central value of forecast. Also, one must take care of wide degree of uncertainty around the central projection. In the early 90's, UK Monetary Policy Committee started to give more emphasis on futuristic view of inflation. According to this recommendation, the Bank of England (BoE), which is the central bank of the UK, published a chart showing Inflation forecast along with the uncertainty measured using forecast errors of previous ten years. But this chart was not able to capture forecast risks and has not included uncertainties around the central projection.

Bank of England first published its Inflation forecasts using Fan charts in Feb 1996. Since that, it has been the essential part of inflation reports of central bank around the world. Fan chart is nothing but the probability distribution of forecast of inflation which mainly takes three parameters central projection, uncertainty and risk surrounding them and gives a way to present the forecast in a pictorial way. Fan chart allows central banks to communicate with a minimum of ambiguity whether the risk is believed to be higher than inflation will be below the forecast than that it will be above. The standard statistical approach in deriving forecast error bands would start with estimating appropriate time series model for making inflation forecasts. The main use of the Fan chart is to improve presentation and focus attention on the whole of the forecast distribution, rather than small changes to the central projection. For policy purposes, having uncertainty bands around the inflation forecast is useful as they serve to illustrate that the inflation forecast is inherently uncertain.

In past 15-20 years, numerous methodologies have been introduced on the derivation of the Fan chart. Trivially, we can assume that the possible outcomes for inflation will symmetrically disperse around the forecast value. This leads to the normality assumption of forecast distribution. But, the upside and downside risks to forecast of the target variable may not be symmetric. This is due to the fact that economic situations leading to both these risks may be totally different and uncertain. That is why it is essential to explore a particular statistical distribution which has a degree of asymmetry in the form of skewness of a variable. Such distributions are plenty in statistics literature. Split Normal, Asymmetric Laplace distributions are regularly used in Econometric or Financial modeling. There is one class of distributions named 'Generalised Hyperbolic' which consists of various distributions of different skewness, kurtosis and shape. In literature, the construction of the Fan chart is based on Two-piece Normal or Split Normal distribution, which is an asymmetric version of usual Gaussian or Normal distribution. There is no work available which uses the other distributions. Hence, this area remains unexplored in the research about the construction of Fan charts.

Many times we cannot study single variable independently, but we need to look the other variables which interact with the study variable. This motivates the researchers for the multidimensional analysis. Although, many researchers across the world have used different approaches for the construction of the Uni-variate Fan chart, there is very limited work done on its multi-dimensional extension. Also, applications of Fan Charts in other fields like Public Health, Reliability were not studied in detail.

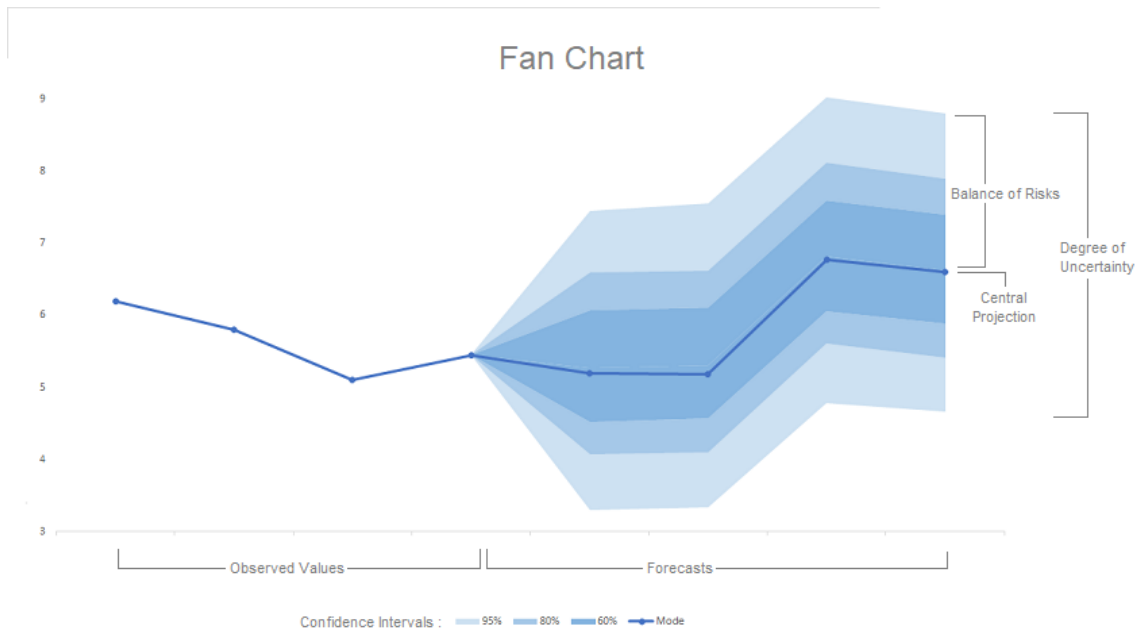


Figure 1.1: *Fan Chart:Illustration*

1.3 India and other BRICS nations

BRICS is the acronym coined for an association of five major emerging national economies: Brazil, Russia, India, China and South Africa - which together represent about 42% of the population, 23% of GDP, 30% of the territory and 18% of the global trade. The prevalence of COVID-19 has affected every single nation in the year 2020. To take into account the effect of this situation on some of the emerging economies in the world, the fancharts for inflation in India and the other BRICS nations are constructed for the period of next six months (Aug 2020 to Jan 2021).

1.4 Overview of the Report

The rest of the report is outlined as follows: All the theory regarding the various probability distributions is discussed in chapter 3. All the relevant derivations are also provided. The Data Analysis part has been described in chapter 4. All the necessary details of various models are given along with the outputs. Also, Fan charts along with the corresponding probability table are given for India as well as other BRICS nations. Finally chapter 5 concludes with some remarks about the work done along with the key findings. Scope and Limitations of the work are also discussed in brief.

Chapter 2

Literature Review

Many central banks around the world are using the Fan Chart methodology in their annual or quarterly inflation bulletin. But, the concept of Fan Chart is not so popular in the academia. Very less research has been done on the construction of Fan charts and its computations. Blix and Sellin (1998) published first detailed article on this topic with emphasis on theoretical background. They attempted to link the inflation uncertainty to the balance of risks for various micro variables. The paper focuses on the derivation of the inflation forecast skewness from uncertainty. All the detailed derivations and expressions about skewness, variances are also provided. The uncertainty in these macro variables is based on their historical standard deviations, but subjective adjustment is also allowed if there is reason to be more or less uncertain than the past. This paper lacks the details on the computations and the model for the inflation.

Román (2007) was the first one to present the details of the Fan chart computation. Authour presents all the technical details along with its implementation. A Visual Basic for Excel program is also available. This implementation provides more flexibility than the previous one since it permits the inputs to be entered on a quarterly basis instead of on a yearly basis. This paper uses Transmission Mechanism Model for the inflation forecast.

The application of Blix and Sellin (1998) in the Indian context was discussed explicitly by Banerjee and Das (2011). They have shown how the balance of risk of any macro-economic indicator can be assessed by the skewness parameter of the distribution. They discussed the use of fan chart in other countries not only in the context of forecasting inflation and growth but also in the study of survival rates. They have used VAR models for modeling the inflation. This is the only work based on the monthly data and perhaps only work in Indian context which deals with univariate Fan charts. This paper would serve as a prime foundation or motivation for our work.

Blix and Sellin (2000) extended their work to bivariate framework. Shaw (2020) also illustrate the uncertainties in forecasts of two mutually dependent variables by using bivariate setup. Using a joint distribution of two related variables, this paper proposes a theory for constructing bivariate fan chart and conditional fan chart for one variable given known information on the other. This is the only available literature regarding bivariate Fan chart in Indian context.

Razi and Loke (2017) suggested alternative methodology for constructing Fan charts using financial market volatility and time-varying coefficients to forecast the inflation in dynamic manner in Malaysia. Cogley et al. (2005) have used Bayesian Fan charts for U.K. inflation. They have estimated a Bayesian vector autoregression with drifting coefficients and stochastic volatilities.

In summary, following points are noted from the Literature survey.

- All the works regarding to Fan Charts have used Split Normal density for the construction. No other asymmetric distribution has been explored.
- Except Banerjee and Das (2011), no other work constructed Fan Chart using Monthly data.
- There are different modeling approaches by every author such as Transmission Mechanism, VAR, BVAR, etc.
- There is not much work done on multidimensional or even bivariate extension of the Fan Chart.

Chapter 3

Theoretical Discussion

3.1 Split Normal Density

The Split Normal distribution which is also known as Two-piece Normal distribution is nothing but asymmetric Normal distribution with different variances for the halves around the mode. This distribution along with its fitting and estimation has been discussed by John (1982) and then by Johnson et al.(1994).

There are several equivalent parameterizations for the split-normal distribution. But, most of the research regarding the Fan charts study only two of them in details.

Let us consider a random variable X . It is said to have a Split Normal or two-piece Normal density if its probability density function is given by,

$$f_X(x, \mu, \sigma_1, \sigma_2) = \begin{cases} C \exp\{\frac{-1}{2\sigma_1^2}(x - \mu)^2\}, & \text{for } -\infty < x \leq \mu \\ C \exp\{\frac{-1}{2\sigma_2^2}(x - \mu)^2\}, & \text{for } \mu < x < \infty \end{cases} \quad (3.1)$$

Where, $-\infty < \mu < \infty$ is the mode, $\sigma_1, \sigma_2 > 0$ are left and right hand side standard deviations respectively and C is a normalizing constant. It can be computed by solving $\int_{-\infty}^{\infty} f_X(x)dx = 1$.

The detailed derivation was given by Banerjee and Das (2012). Hence, C is given by $C = \sqrt{\frac{2}{\pi}}(\sigma_1 + \sigma_2)^{-1}$.

If $\sigma_1 = \sigma_2$, we get our usual Normal p.d.f.

It is noted that the above density is completely specified by the triplet $(\mu, \sigma_1, \sigma_2)$.

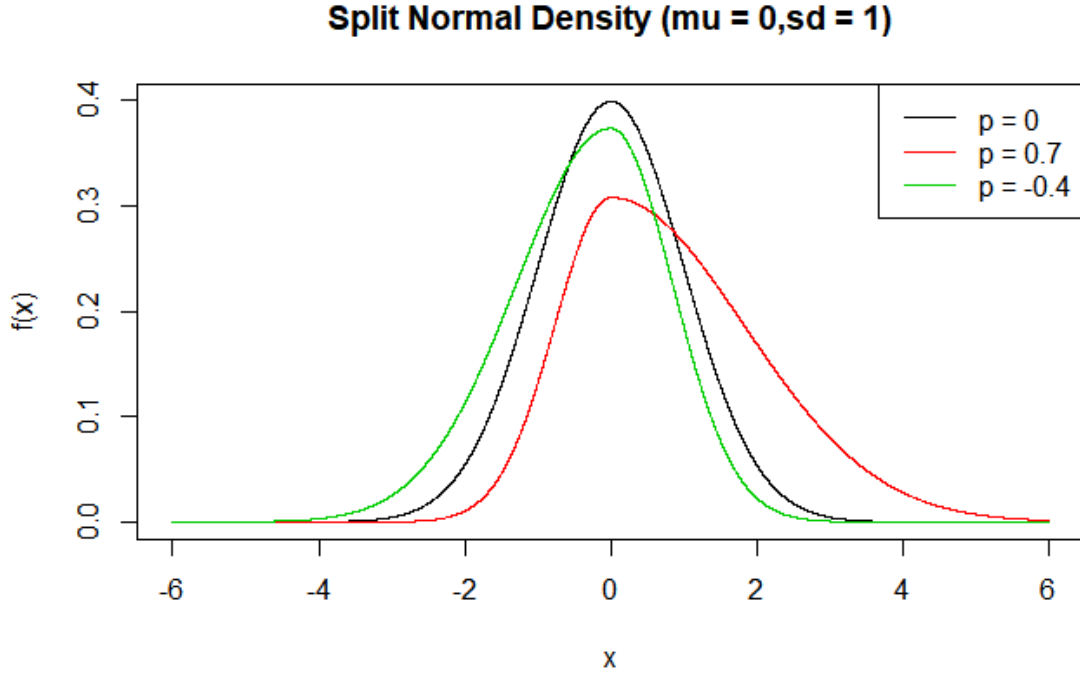


Figure 3.1: *Illustration of Split Normal with different skewness parameter*

The mean of the distribution is given by,

$$\tilde{\mu} = \int_{-\infty}^{\infty} x f_X(x) dx \quad (3.2)$$

$$= \int_{-\infty}^{\mu} x f_X(x) dx + \int_{\mu}^{\infty} x f_X(x) dx \quad (3.3)$$

$$= I_1 + I_2 \quad (3.4)$$

Solving the two integrals I_1 and I_2 , we get

$$I_1 = \mu \frac{\sigma_1}{\sigma_1 + \sigma_2} - \sqrt{\frac{2}{\pi}} \frac{\sigma_1^2}{\sigma_1 + \sigma_2} \quad (3.5)$$

$$I_2 = \mu \frac{\sigma_2}{\sigma_1 + \sigma_2} + \sqrt{\frac{2}{\pi}} \frac{\sigma_2^2}{\sigma_1 + \sigma_2} \quad (3.6)$$

Therefore,

$$\tilde{\mu} = I_1 + I_2 \quad (3.7)$$

$$= \sqrt{\frac{2}{\pi}}(\sigma_2 - \sigma_1) + \mu \quad (3.8)$$

The variance of the distribution is,

$$V[X] = (1 - \frac{2}{\pi})(\sigma_2 - \sigma_1)^2 + \sigma_1\sigma_2 \quad (3.9)$$

The skewness indicator is given by,

$$\xi = \tilde{\mu} - \mu \quad (3.10)$$

$$= \sqrt{\frac{2}{\pi}}(\sigma_2 - \sigma_1) \quad (3.11)$$

When $\xi \leq 0$, the distribution is biased to the left and when $\xi \geq 0$ then the distribution is biased to the right.

The balance of risk is determined by the skewness around the central projection. If the balance of risk is on the upside, the mean will be greater than the mode and if it is on the downside, the mean will be smaller than the mode. The balance of risk is given by,

$$\rho = P(X \leq \mu) = \frac{\sigma_1}{\sigma_1 + \sigma_2}. \quad (3.12)$$

We can compute,

$$P[X \leq k] = \begin{cases} C \sqrt{2\pi}\sigma_1\Phi(\frac{k-\mu}{\sigma_1}), & \text{for } k \leq \mu \\ 1 - C \sqrt{2\pi}\sigma_2[1 - \Phi(\frac{k-\mu}{\sigma_2})], & \text{for } k > \mu \end{cases} \quad (3.13)$$

Where, $\Phi(\cdot)$ is the CDF of standard Normal distribution.

Quantiles of the distributions are given by,

$$k = \begin{cases} \mu + \sigma_1\Phi^{-1}(\frac{\alpha}{C\sqrt{2\pi}\sigma_1}), & \text{for } \alpha \leq p \\ \mu + \sigma_2\Phi^{-1}(\frac{\alpha+C\sqrt{2\pi}\sigma_2-1}{C\sqrt{2\pi}\sigma_2}), & \text{for } \alpha > p \end{cases} \quad (3.14)$$

Where, $\alpha = P[X \leq \mu]$

We have an alternative parametrization of the above distribution which takes into account three parameters regarding mode(μ), uncertainty(σ) and skewness(γ). This can be found in Johnson, Kotz and Balakrishnan (1994). The p.d.f. is given by,

$$f_X(x, \mu, \sigma, \gamma) = \begin{cases} \frac{A}{\sqrt{2\pi\sigma}} \exp\left\{-\frac{1-\gamma}{2\sigma^2}[(x-\mu)^2]\right\} & \text{if } x \leq \mu \\ \frac{A}{\sqrt{2\pi\sigma}} \exp\left\{-\frac{1+\gamma}{2\sigma^2}[(x-\mu)^2]\right\} & \text{if } x > \mu \end{cases} \quad (3.15)$$

where, $A = \frac{2}{\frac{1}{\sqrt{1-\gamma}} + \frac{1}{\sqrt{1+\gamma}}}$ and $-1 < \gamma < 1$.

If we compare the two parametrizations i.e. equations 3.1 and 3.14, we get

$$\sigma_1 = \sqrt{\frac{\sigma^2}{1-\gamma}} \quad (3.16)$$

$$\sigma_2 = \sqrt{\frac{\sigma^2}{1+\gamma}} \quad (3.17)$$

Here, Uncertainty (σ) can be calculated as average over historical standard deviations.

If we substitute σ_1 and σ_2 in 3.11,

$$\frac{1}{\rho} = 1 + \sqrt{\frac{1-\gamma}{1+\gamma}} \quad (3.18)$$

$$\Rightarrow \gamma = \frac{2\rho - 1}{1 - 2\rho + 2\rho^2} \quad (3.19)$$

And finally, from 3.10, 3.15 and 3.16,

$$\gamma = \begin{cases} -\sqrt{1 - \left(\frac{\sqrt{1+2\beta}-1}{\beta}\right)^2} & \text{if } \xi > 0 \\ \sqrt{1 - \left(\frac{\sqrt{1+2\beta}-1}{\beta}\right)^2} & \text{if } \xi < 0 \end{cases} \quad (3.20)$$

where, $\beta = \frac{\pi}{\sigma^2} \xi^2$

Thus, fan chart can be drawn if we know the triplet $(\mu, \sigma_1, \sigma_2)$.

3.2 Generalised Hyperbolic Distribution

The Generalized Hyperbolic Distribution (GHD) family was originally discussed by Barndorf and Nielsen (1977, 1978). It is the family of continuous type distributions

such that the logarithm of the probability density function is a hyperbola. It has been showed that this distribution arose from specific normal mean-variance mixture. The class of generalized hyperbolic distributions and its sub classes – the hyperbolic and the normal inverse Gaussian distributions – possess the property of having semi heavy tails, which is of interest for financial or econometric modeling.

The pdf for generalised hyperbolic distributions is given by :

$$f(x; \lambda, \alpha, \beta, \delta, \mu) = a_\lambda (\delta^2 + (x - \mu)^2)^{(\lambda-1/2)/2} K_{\lambda-1/2} * (\alpha \sqrt{(\delta^2 + (x - \mu)^2)}) \exp(\beta(x - \mu)) \quad (3.21)$$

where , $a_\lambda = \frac{(\alpha^2 - \beta^2)^{\lambda/2}}{2\pi\alpha^{\lambda-1/2}\delta^\lambda K_\lambda \delta \sqrt{(\alpha^2 - \beta^2)}}$; $(x, \mu \in R)$

$$\begin{aligned} \delta &\geq 0, |\beta| < \alpha \text{ if } \lambda > 0 \\ \delta &> 0, |\beta| < \alpha \text{ if } \lambda = 0 \\ \delta &> 0, |\beta| \leq \alpha \text{ if } \lambda < 0 \end{aligned}$$

Here, μ is scale, δ is location, β is asymmetry, α is shape and λ is kurtosis parameter.

K_λ is a modified Bessel function, which are canonical solutions $y(x)$ of Bessel's differential equation, which is given by,

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - \lambda^2)y = 0 \quad (3.22)$$

Where λ is an arbitrary complex number.

Note that, the normal distribution can be obtained as a limiting case of generalised hyperbolic distribution for $\delta \rightarrow \infty$ and $\frac{\delta}{\alpha} \rightarrow \sigma^2$.

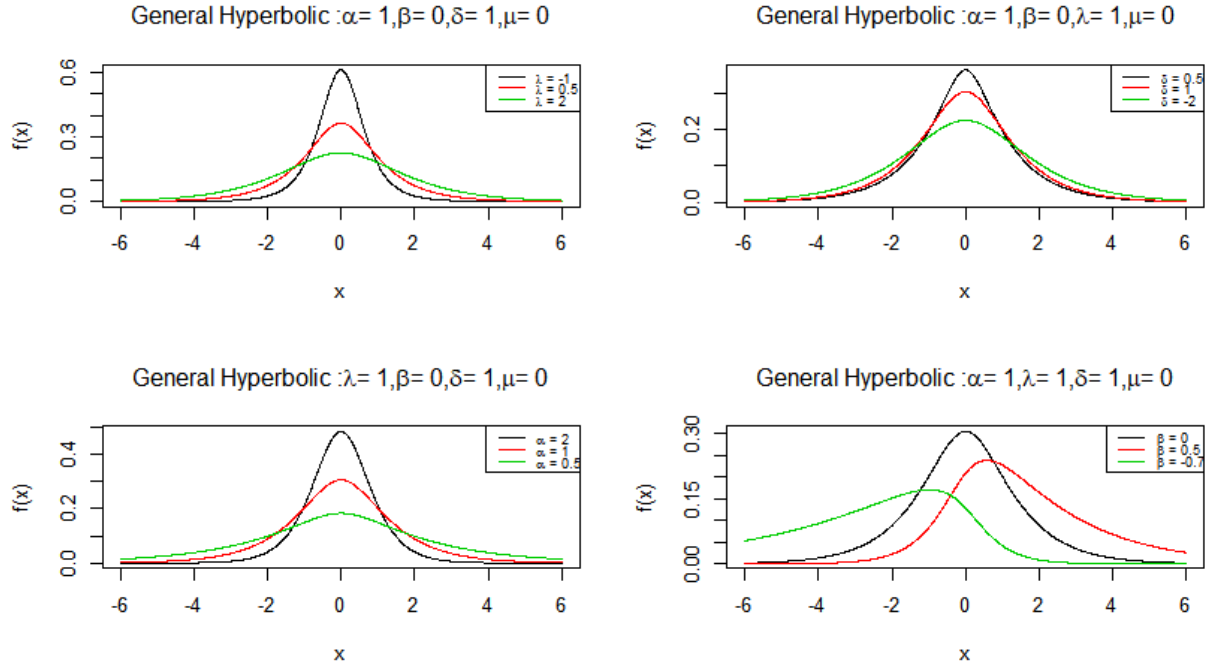


Figure 3.2: Illustration of Generalised Hyperbolic with different parameters

The computations are mean and variances of generalised hyperbolic distributions are tedious, hence, they are omitted from this discussion. Karsten Prause (1999) give all these details along with some more discussion about the properties of the distribution.

We can write Log Likelihood for p.d.f. given in 3.24 as,

$$\begin{aligned} \log L &= \log a(\lambda, \alpha, \beta, \delta) + \left(\frac{\lambda}{2} - \frac{1}{4}\right) \sum_{i=1}^n \log(\delta^2 + (x_i - \mu)^2) \\ &\quad + \sum_{i=1}^n [\log K_{\lambda-\frac{1}{2}}(\alpha\sqrt{\delta^2 + (x_i - \mu)^2}) + \beta(x_i - \mu)] \end{aligned} \quad (3.23)$$

Where, we assume the independent of observations $x_i, i = 1, 2, \dots, n$ and a is given by,

$$a(\lambda, \alpha, \beta, \delta) = \frac{(\alpha^2 - \beta^2)^{\lambda^2}}{\sqrt{2\pi}\alpha^{\lambda-\frac{1}{2}}\delta^{\lambda}K_{\lambda}(\delta\sqrt{\alpha^2 - \beta^2})} \quad (3.24)$$

we obtain direct solutions of the likelihood equations for μ and β by solving $\frac{d}{d\mu} \log L = 0$ and $\frac{d}{d\beta} \log L = 0$. However, we need to have some numerical estimation method to obtain the solutions for other parameters. R package 'fBasics' uses `nlm()` function to minimize the "negative" maximum log-likelihood function. `nlm` carries out a minimization using a Newton-type algorithm. This method is discussed by Dennis and Schnabel

(1983).

Note that, there are huge number of distributions that can be seen as the special cases of generalised hyperbolic distribution. Below are the some of special cases of the generalised hyperbolic distribution,

1. $\lambda = 1$: Hyperbolic distribution
2. $\lambda = \frac{-1}{2}$: Normal-Inverse Gaussian
3. $\delta = 0$: Variance-Gamma distribution

Generalized hyperbolic distributions and their subclasses have been proposed as a model for the distribution of increments of financial price processes.(Eberlein/Keller 1995),(Ryder 1996)

Chapter 4

Data Analysis

We have the data of Consumer Price Index (CPI) for India from Jan 2011 to Oct 2019.

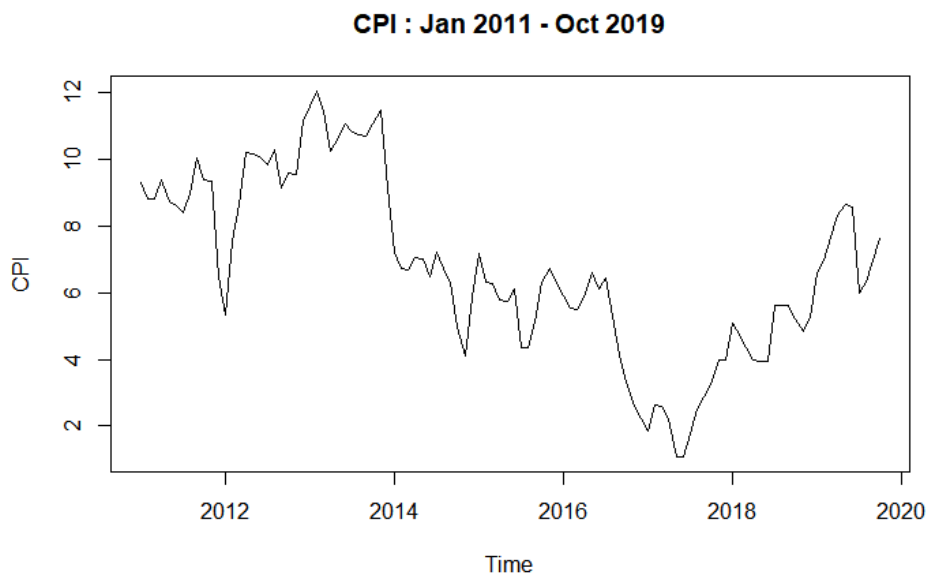


Figure 4.1: *CPI (%)*

4.1 Data Description

1. **Consumer Price Index (CPI):** It measures changes in the price level of a weighted average market basket of consumer goods and services purchased by households. The CPI is a statistical estimate constructed using the prices of a sample of representative items whose prices are collected periodically.

Source: Organization for Economic Co-operation and Development, Consumer Price Index: All Items for India [INDCPIALLMINMEI], retrieved from FRED,

Federal Reserve Bank of St. Louis;

<https://fred.stlouisfed.org/series/INDCPIALLMINMEI>, March 9, 2020

2. **Real effective exchange rate (REER):** It is the weighted average of a country's currency in relation to an index or basket of other major currencies. The weights are determined by comparing the relative trade balance of a country's currency against each country within the index.

Source: Bank for International Settlements, Real Broad Effective Exchange Rate for India [RBINBIS], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/RBINBIS>, March 9, 2020.

3. **Economic Policy Uncertainty Index:** A new index of economic policy uncertainty (EPU) developed by Baker, Bloom and Davis (2015).

Source: Baker, Scott R., Bloom, Nick and Davis, Stephen J., Economic Policy Uncertainty Index for India [INDEPUINDXM], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/INDEPUINDXM>, March 8, 2020.

4. **M3 Indicator:** M3 is a measure of the money supply that includes M2 as well as large time deposits, institutional money market funds, short-term repurchase agreements and larger liquid assets. M3 can be thought of as a congregation of all the other classifications of money (M0, M1 and M2) plus all of the less liquid components of the money supply.

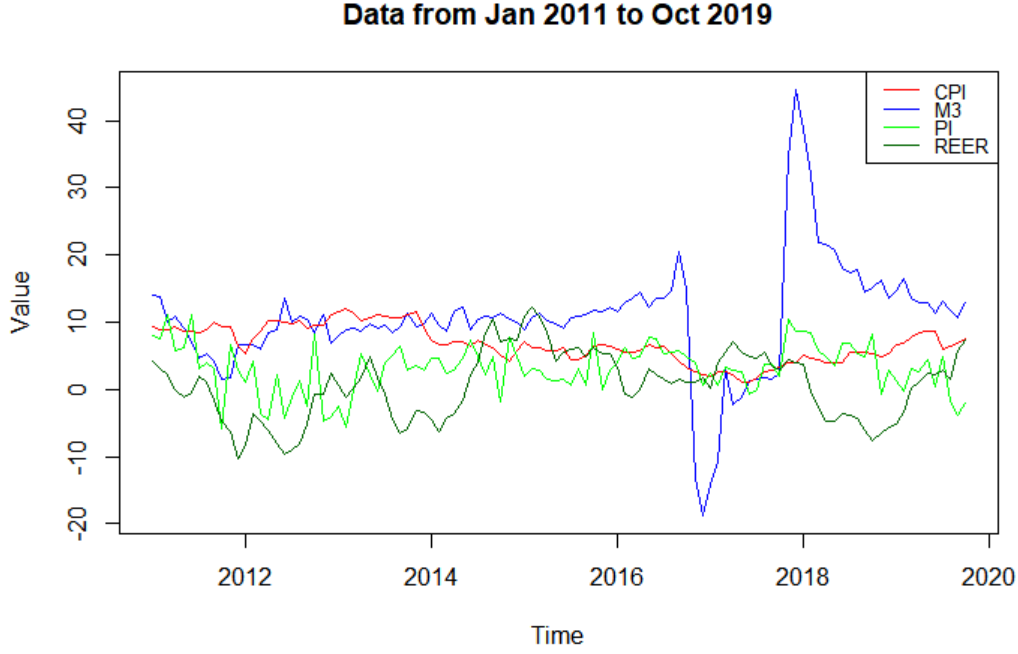
Source: Organization for Economic Co-operation and Development, M3 for India [MABMM301INM189S], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/MABMM301INM189S>, March 9, 2020.

5. **Industrial Production:** The Index of Industrial Production (IIP) is an index which shows the growth rates in different industry groups of the economy in a stipulated period of time. The IIP index is computed and published by the Central Statistical Organisation (CSO) on a monthly basis.

Source: Organization for Economic Co-operation and Development, Production of Total Industry in India [INDPROINDMISMEI], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/INDPROINDMISMEI>, March 8, 2020.

All the data have monthly frequency and base year as 2015.

The data for all the variables mentioned above (excluding Uncertainty) for India from Jan 2011 to Oct 2019.

**Figure 4.2:** *Data*

4.2 ARIMA

Autoregressive (AR) models are based on the idea that the current value of the series, x_t , can be explained as a function of p past values, $x_{t1}, x_{t2}, \dots, x_{tp}$ where p determines the number of steps into the past needed to forecast the current value. The moving average (MA) models assume the white noise w_t on the right-hand side of the defining equation are combined linearly to form the observed data. An ARMA model with parameters (p, q) is given by :

$$x_t = \phi_1 x_{t-1} + \dots + \phi_p x_{t-p} + \omega_t + \theta_1 \omega_{t-1} + \dots + \theta_q \omega_{t-q} \quad (4.1)$$

where $\phi_p \neq 0$, $\theta_q \neq 0$ and $\sigma_\omega^2 > 0$.

The integrated ARMA, or ARIMA, model is a broadening of the class of ARMA models to include differencing.

A process x_t is ARIMA(p, d, q) if $\Delta^d x_t$ is ARMA(p, q), where Δ denotes the difference component.

A seasonal autoregressive integrated moving average model, or SARIMA model denoted by SARIMA(P, D, Q)(p, d, q)[S] is given by

$$\Phi_P(B^s)\phi(B)\Delta_s^D\Delta^d x_t = \Theta_Q(B^s)\theta(B)\omega_t \quad (4.2)$$

The ordinary autoregressive and moving average components are represented by polynomials $\phi(B)$ and $\theta(B)$ of orders p and q , respectively, and the seasonal autoregressive and moving average components by $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ of orders P and Q and ordinary and seasonal difference components by $\Delta^d = (1 - B)^d$ and $\Delta^D = (1 - B^s)^D$.

ARIMA and SARIMA models are fit to the data by observing ACF and PACF. AICc is used as the basis for the selection of the best fit model. A model is developed using the data till Oct-19 and forecasts have been made for next six months.

Details about the models compared are given below.

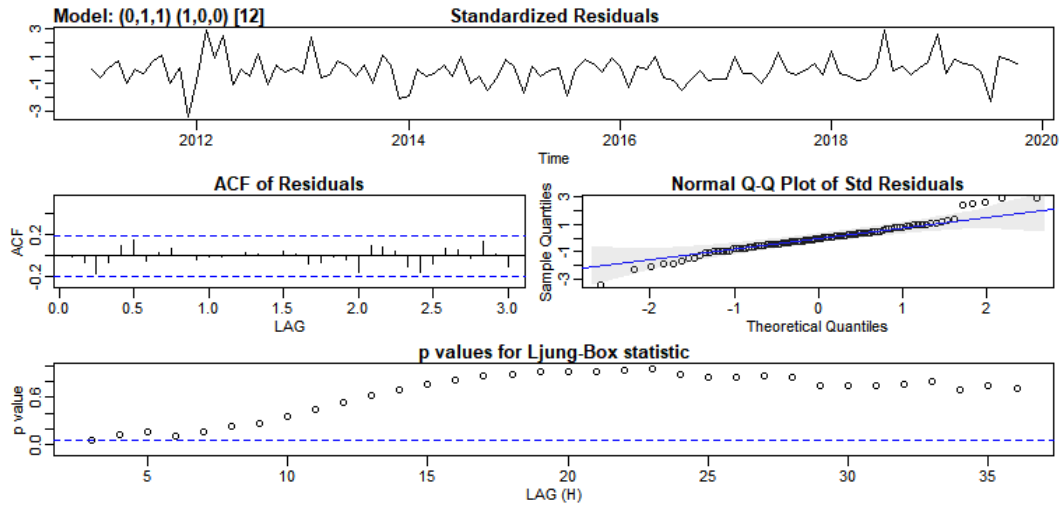
Model	AICc	BIC	Log Likelihood
ARIMA(1,1,0)	266.237	271.422	-131.057
ARIMA(1,1,1)	267.493	275.217	-130.628
SARIMA(1,1,0)(1,0,0)[12]	239.348	247.072	-116.555
SARIMA(0,1,1)(1,0,0)[12]	238.729	246.453	-116.246

SARIMA(0,1,1)(1,0,0)[12] model is found to be most suitable using AICc and BIC. Both the MA and seasonal AR coefficient seem to be significant. SARIMA modeling of Inflation data has been used by Živko and Bošnjak (2017) for the Inflation in Croatia. Hence, fitted model is given by, The model is given by :

$$X_t = X_{t-1} - 0.5409X_{t-12} + 0.2102\epsilon_{t-1} + \epsilon_t \quad (4.3)$$

4.2.1 Residual Diagnostics

The following graphs show some of the residual diagnostics carried out.



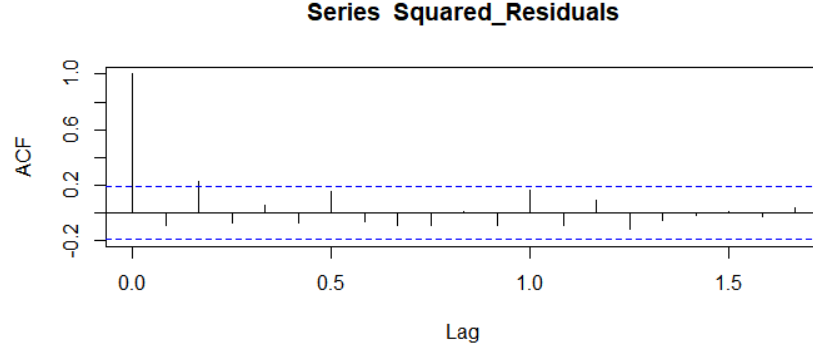


Figure 4.3: *Residual Diagnostics*

The standardised residuals of the SARIMA(0,1,1)(1,0,0)[12] model show no specific pattern. ACF shows all values contained between the bounds, hence the residuals are uncorrelated. Residuals are normally distributed. The p values for all the lags for Ljung-Box statistic are high. The ACF of the squared residuals is also contained between the bands. Hence, there is no evidence of ARCH or GARCH effect.

Hence we say the residuals are white noise. The model SARIMA(0,1,1)(1,0,0)[12] is acceptable for the given series.

4.3 Vector Auto-Regressive Model

VAR models (vector autoregressive models) are used for multivariate time series. The structure is that each variable is a linear function of past lags of itself and past lags of the other variables.

In general, for a VAR(p) model, the first p lags of each variable in the system would be used as regression predictors for each variable.

X_t is a VAR(p) process if X_t is stationary and for every t ,

$$X_t - \Phi_1 X_{t-1} - \dots - \Phi_p X_{t-p} = Z_t \quad (4.4)$$

$$\Phi(B)X_t = Z_t \quad (4.5)$$

where, Z_t is WN(0, Σ), Φ is a matrix valued polynomial.

The data from Jan 2011 to Oct 2019 for the variables : CPI, Policy Uncertainty, M3, IIP and REER is used to train the model. The VAR models with different lags are compared on the basis of AIC.

Lag	1	2	3	4	5
AIC	14.567	14.390	14.304	14.399	14.493

According to the AIC, VAR of order 3 is found to be most suitable.

The model is given by :

$$\begin{aligned} CPI_t = & 1.1403 \, CPI_{t-1} + 0.0029 \, Uncertainty_{t-1} + 0.0262 \, M3_{t-1} - 0.0390 \, PI_{t-1} + \\ & 0.0042 \, REER_{t-1} - 0.4048 \, CPI_{t-2} - 0.0018 \, Uncertainty_{t-2} - 0.0046 \, M3_{t-2} + 0.0014 \\ & PI_{t-2} - 0.0090 \, REER_{t-2} + 0.1793 \, CPI_{t-3} - 0.0019 \, Uncertainty_{t-3} - 0.0128 \, M3_{t-3} - \\ & 0.0040 \, PI_{t-3} - 0.0252 \, REER_{t-3} + 0.6156 \end{aligned}$$

4.3.1 Residual Diagnostics

The following graphs show some of the residual diagnostics carried out.

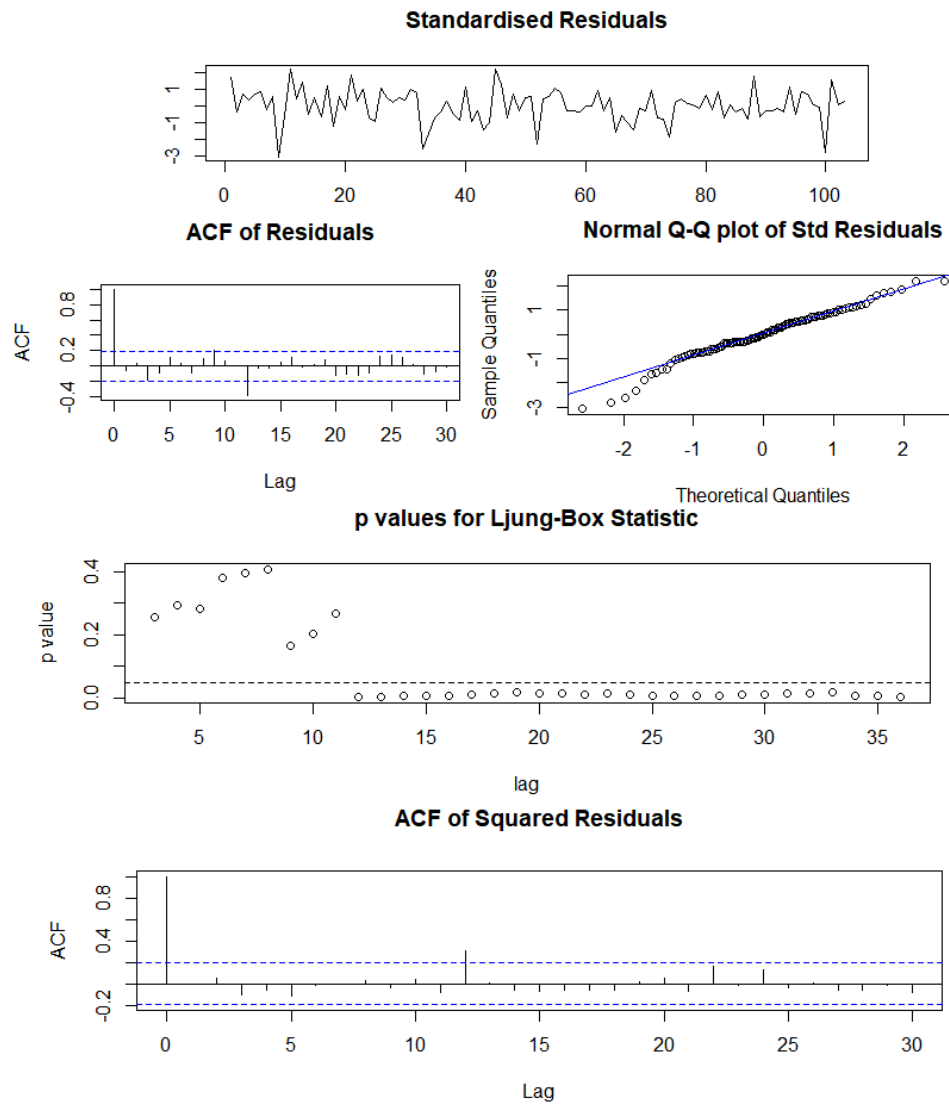


Figure 4.4: *Residual Diagnostics*

The standardised residuals of the VAR(3) model show no specific pattern. ACF shows all but one values contained between the bounds, hence the residuals are uncorrelated. Residuals deviate slightly from normal distribution. The p values for all the lags for Ljung-Box statistic are small for higher lags. The ACF of the squared residuals is contained between the bands. Hence, no ARCH or GARCH effect is observed.

Hence there is no enough evidence of residues being white noise. This indicates that the model VAR(3) is not a proper choice of time series model.

4.3.2 Test for Cointegration

Cointegration occurs when two or more nonstationary time series have a long run equilibrium and they tend to move together in such a way that their linear combination results in a stationary time series.

Johansen's test is a way to determine if three or more time series are cointegrated. More specifically, it assesses the validity of a cointegrating relationship, using a maximum likelihood estimates (MLE) approach. Test for cointegration using Johansen's procedure rejects the cointegration between the variables and ΔX is a stable VAR(2) process.

4.4 SETAR Model

The Threshold Autoregression (TAR) model is an autoregression allowing for two or more branches governed by the values for a threshold variable. This allows for asymmetric behavior that cannot be explained by a single ARMA model. A two branch model is given by :

$$X_t = \begin{cases} \alpha^{(1)} + \phi_{11}X_{t-1} + \dots + \phi_{1p}X_{t-p} & \text{if } Z_{t-d} \leq \gamma \\ \alpha^{(2)} + \phi_{21}X_{t-1} + \dots + \phi_{2q}X_{t-q} & \text{if } Z_{t-d} > \gamma \end{cases} \quad (4.6)$$

A SETAR model (Self-Exciting TAR) is a special case where the threshold variable is the variable itself. The model with one threshold is given by :

$$X_t = \begin{cases} \alpha^{(1)} + \phi_{11}X_{t-1} + \dots + \phi_{1p}X_{t-p} & \text{if } X_{t-d} \leq \gamma \\ \alpha^{(2)} + \phi_{21}X_{t-1} + \dots + \phi_{2q}X_{t-q} & \text{if } X_{t-d} > \gamma \end{cases} \quad (4.7)$$

The best model is determined by comparing the pooled AIC for different combinations of autoregressive orders for higher and lower regime with 1 threshold. The results for the grid search for 1 threshold (First 10 models with minimum pooled AIC) are given below.

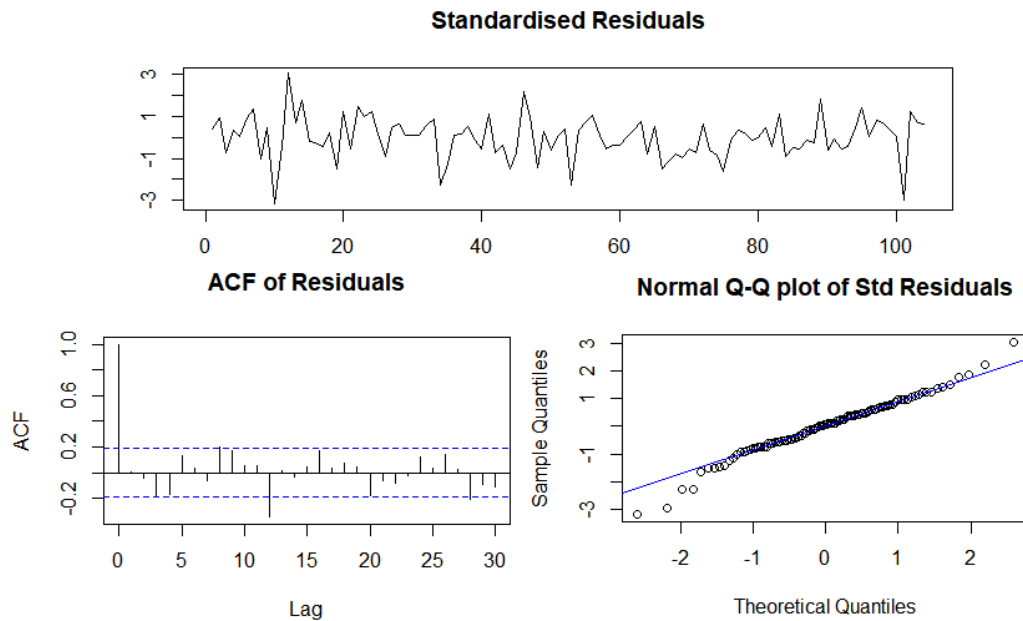
AR order for Lower Regime	AR order for Higher Regime	Threshold	Pooled AIC
2	1	9.411765	265.4101
2	1	9.340659	266.3470
2	1	8.426966	266.5427
2	1	9.392265	266.6041
2	1	9.132420	266.6278
2	1	8.333333	266.7105
2	2	7.239819	266.8031
2	2	8.426966	266.8125
2	2	8.333333	266.9306
2	1	9.547739	267.0019

Hence the model with AR order 2 for Lower regime and AR order 1 for Higher regime, with threshold 9.412 is selected. The model is given by :

$$X_t = \begin{cases} 0.4954 + 1.1939X_{t-1} - 0.2730X_{t-2} & \text{if } X_{t-1} \leq 9.412 \\ 4.8169 + 0.5371X_{t-1} & \text{if } X_{t-1} > 9.412 \end{cases} \quad (4.8)$$

4.4.1 Residual Diagnostics

The following graphs show some of the residual diagnostics carried out.



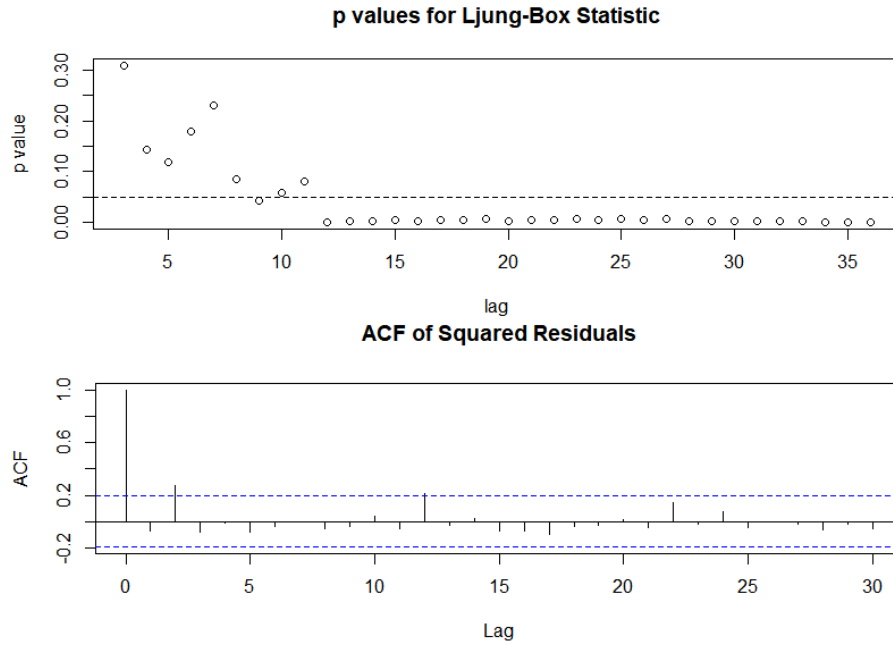


Figure 4.5: *Residual Diagnostics*

The standardised residuals of the SETAR model show no specific pattern. ACF shows all but one values contained between the bounds, hence the residuals are uncorrelated. Residuals are normally distributed. The p values for all the lags for Ljung-Box statistic are small for higher lags. The ACF of the squared residuals is also contained between the bands. Hence, there is no evidence of ARCH or GARCH effect.

Hence we say the residuals are not white noise. The model SETAR is not a good model for the given series.

4.5 Model Selection

For the Inflation data, three different modeling approaches are tried. One is using popular ARIMA model, where we have made use of the seasonal component of the data to fit a seasonal ARIMA or SARIMA model. A multivariate version of AR model i.e. VAR model has also been used with the consideration of other macroeconomic indicators along with the CPI. Also, we took into account a policy uncertainty by adding its index as one of the variables. Third approach is totally different from first two and non-traditional. We have used SETAR, a non-linear time series model for the Univariate Inflation data. This model is quite useful when there are different behavioural patterns in a single time series.

Model selection is always depends upon your objective and motivation. Also, it has lot to do with the interpretability. We restrict ourselves only with the accuracy of the

model. Hence, model with best test set accuracy is considered as the desired model. Five different metrics for test set accuracy have been used. The details of those are as follows:

$x_i = i^{th}$ actual observation, $\hat{x}_i = i^{th}$ forecast, h = Forecast horizon

1. ME (Mean Error) : Arithmetic Mean of the Errors i.e. $\frac{1}{h} \sum_{i=1}^h (x_i - \hat{x}_i)$
2. RMSE (Root Mean Squared Error): The standard deviation of the residuals (prediction errors).

$$RMSE = \sqrt{\frac{1}{h} \sum_{i=1}^h (x_i - \hat{x}_i)^2}$$

3. MAE (Mean Absolute Error) : Arithmetic mean of absolute errors. i.e. $\frac{1}{h} \sum_{i=1}^h |(x_i - \hat{x}_i)|$
4. MPE (Mean Percentage Error) : Average of percentage errors by which forecasts of a model differ from actual values of the quantity being forecast.

$$MPE = \frac{100\%}{h} \sum_{i=1}^h \frac{(x_i - \hat{x}_i)}{x_i}$$

5. MAPE (Mean Absolute Percentage Error): Average of percentage of absolute errors by which forecasts of a model differ from actual values of the quantity being forecast.

$$MAPE = \frac{100\%}{h} \sum_{i=1}^h \left| \frac{(x_i - \hat{x}_i)}{x_i} \right|$$

From Table 5.1, we can see that the except ME, all other metrics have lowest value for SARIMA model. Hence, we choose SARIMA as our best model

	ME	RMSE	MAE	MPE	MAPE
SARIMA	0.3193909	0.9766388	0.7916732	1.95045	10.56814
VAR	-0.1360575	1.306223	1.094137	-5.818319	16.14455
SETAR	0.1404915	1.347737	1.140652	2.361557	15.49363

Table 4.1: *Model-wise Accuracy*

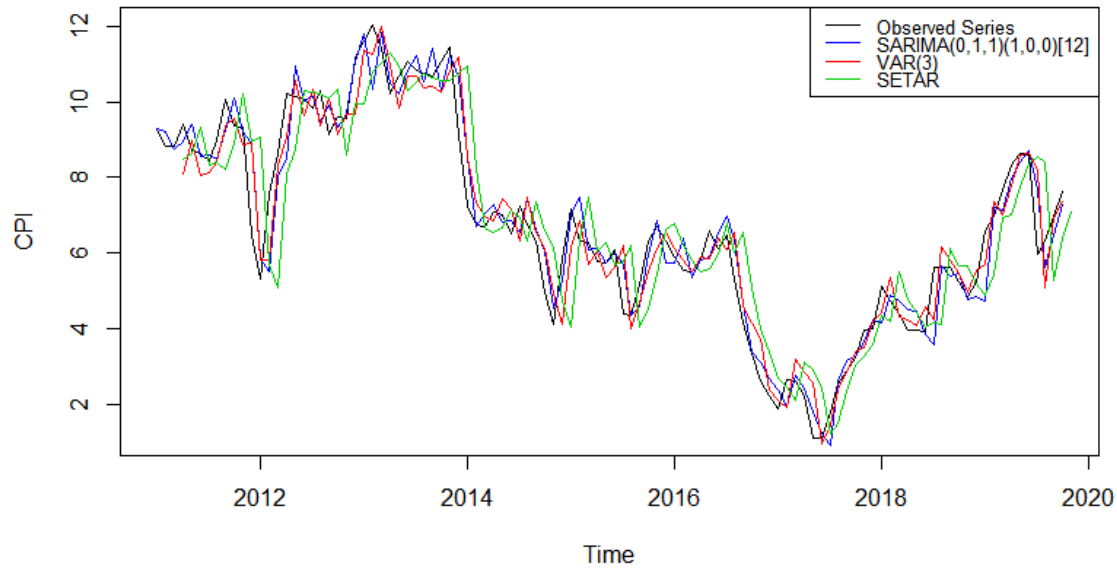


Figure 4.6: *Observed Series with fitted values for each model*

4.6 Construction of Fan Charts

Following are the steps for constructing the Inflation Fan Chart:

1. Fit an appropriate SARIMA model to the data.
2. Using the above model, generate the forecasts for next six months.
3. Fit Split Normal or Generalised Hyperbolic distribution to the forecasts by using estimating its parameters by numerical methods.
4. Compute various percentiles of the distribution required for constructing the Fan chart.
5. Construct the Fan chart using the Mode forecast and above percentiles.

4.6.1 India : Observed Data

An R program is developed for computing the probabilities and construction of Fan Charts. The Probability distribution of six-month forecast is given below.

CPI	Nov-19	Dec-19	Jan-20	Feb-20	Mar-20	Apr-20
<4%	0.1	0.0	0.6	0.8	1.4	3.7
4-4.5%	0.1	0.2	1.0	1.1	2.9	5.5
4.5-5%	0.5	0.1	2.3	2.8	5.7	9.9
5-5.5%	1.1	1.7	5.4	7.7	11.5	15.9
5.5-6%	2.7	3.8	11.2	13.9	18.0	19.2
6-6.5%	5.5	7.6	17.6	18.2	21.4	19.7
6.5-7%	12.0	13.1	20.6	21.4	16.5	13.0
7-7.5%	15.8	19.2	18.7	16.5	12.5	6.1
7.5-8%	22.8	21.0	10.7	8.2	5.8	4.0
8-8.5%	16.4	16.6	7.3	5.3	2.3	2.3
8.5-9%	11.8	10.0	2.7	2.4	1.6	0.5
9-9.5%	6.7	3.4	1.1	0.9	0.3	0.1
9.5-10%	2.4	1.6	0.6	0.6	0.1	0.1
>10%	2.1	1.7	0.2	0.2	0.0	0.0
Actual	8.61	9.63	7.49	6.84	5.5	5.45
Prob[<Mode]	48.4	47.9	46.5	58.7	48.0	52.8

Table 4.2: Probability Table (in %) - Nov 19 to Apr 20 (SARIMA-SND)

Fan Chart from above information is constructed using MS Excel.

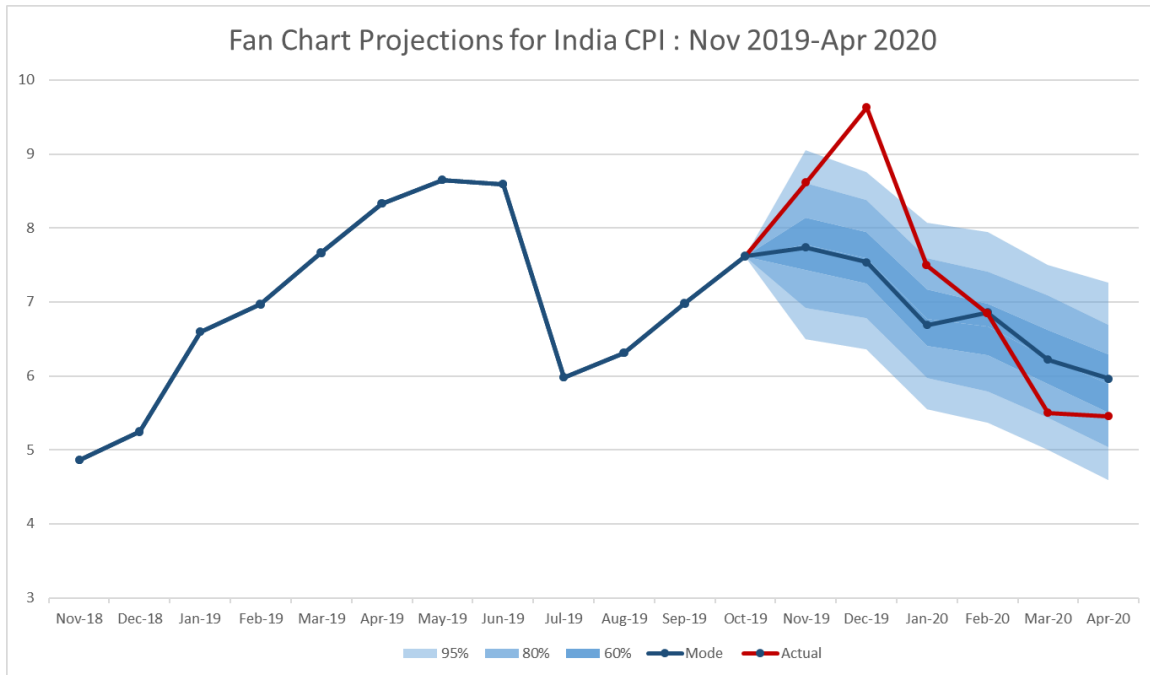


Figure 4.7: India CPI Nov 19 - Apr 20 : Split Normal Distribution

CPI	Nov-19	Dec-19	Jan-20	Feb-20	Mar-20	Apr-20
<4%	0.0	0.1	0.4	0.8	1.9	2.3
4-4.5%	0.0	0.5	0.7	1.1	1.8	4.5
4.5-5%	0.4	0.6	2.7	2.9	4.8	7.6
5-5.5%	0.8	2.2	3.8	6.4	9.7	16.0
5.5-6%	2.1	2.6	9.3	11.2	16.3	21.4
6-6.5%	5.8	5.4	16.6	18.8	23.0	22.1
6.5-7%	9.5	11.3	25.3	21.6	20.4	12.9
7-7.5%	18.3	22.9	20.0	16.0	12.0	5.8
7.5-8%	23.4	22.2	10.4	9.7	5.3	3.8
8-8.5%	19.6	16.5	4.9	5.9	2.9	1.9
8.5-9%	9.9	7.6	2.9	2.7	1.1	1.5
9-9.5%	5.1	4.5	2.1	1.4	0.4	0.0
9.5-10%	2.7	1.6	0.5	0.8	0.2	0.2
>10%	2.4	2.0	0.4	0.7	0.2	0.0
Actual	8.61	9.63	7.49	6.84	5.5	5.45
Prob[<Mode]	52.2	44.6	49.4	48.5	53.9	49.7

Table 4.3: Probability Table (in %) - Nov 19 to Feb 20 (SARIMA - GH)

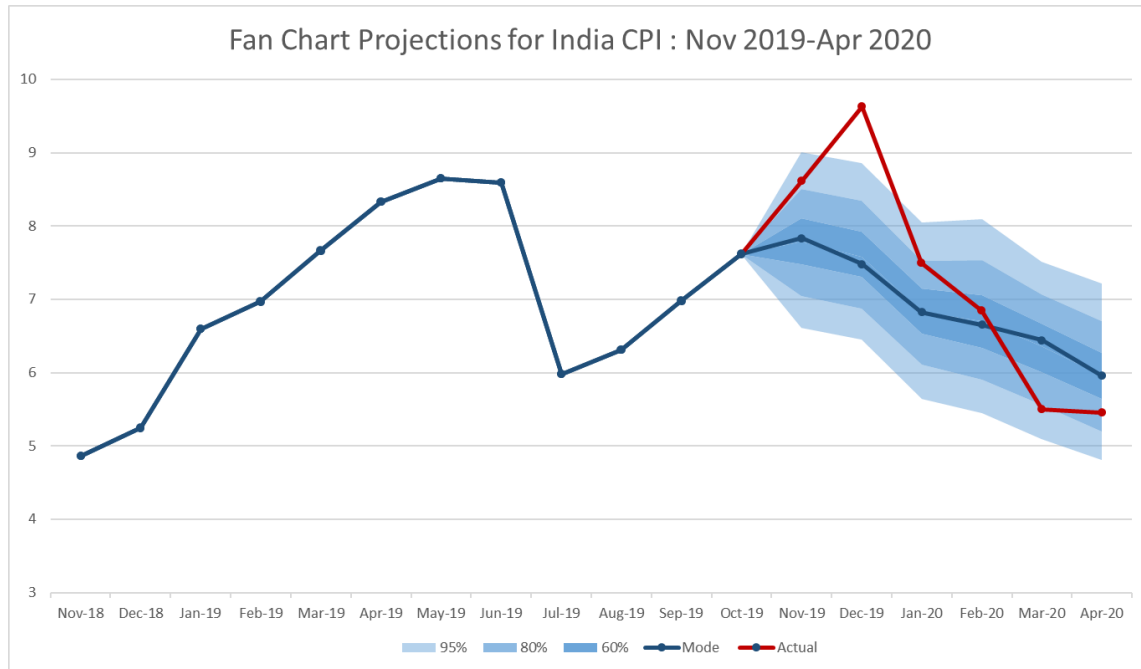


Figure 4.8: India CPI Nov 19 - Apr 20 : Generalised Hyperbolic Distribution

According to the Fan Charts based on Split Normal density upward risks to the forecast would have remained higher than that of downward risks for all the months except February and April 2020. Fan chart constructed using Generalised Hyperbolic distribution indicates that upward risks would be higher for all the months except November 2019 and March 2020. Both of these Fan charts show that actual CPI for the corresponding months lie close to the most likely class. With the exception of December 2019, all the actual observed CPI lie within the confidence bands proposed.

Month	Conf Level	Split Normal Distribution	Generalised Hyperbolic Distribution	Gain in Efficiency
Nov-19	30%	0.7139	0.6362	0.0777
	60%	1.6885	1.4560	0.2325
	80%	2.5501	2.3958	0.1543
Dec-19	30%	0.7004	0.6211	0.0793
	60%	1.5970	1.4732	0.1238
	80%	2.3939	2.4101	-0.0162
Jan-20	30%	0.7615	0.6183	0.1432
	60%	1.6135	1.4220	0.1915
	80%	2.5306	2.4044	0.1262
Feb-20	30%	0.7028	0.7156	-0.0128
	60%	1.6234	1.6280	-0.0046
	80%	2.5704	2.6449	-0.0745
Mar-20	30%	0.7300	0.6540	0.0760
	60%	1.6477	1.5089	0.1388
	80%	2.5040	2.4232	0.0808
Apr-20	30%	0.7874	0.6307	0.1567
	60%	1.6529	1.4977	0.1552
	80%	2.6649	2.3959	0.2690

Table 4.4: *Comparison of Widths - India: Nov 19 to Apr 20*

Except for the month of February, the confidence bands using GHD are narrower than that of SND for almost all the intervals. This shows that GHD is more efficient in terms of reducing the widths of the confidence bands.

4.6.2 India: Future Inflation

An R program is developed for computing the probabilities and construction of Fan Charts using the data on CPI for India for the period Jan 2011 to July 2020. The Probability distribution of six-month forecast is given below.

CPI	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20	Jan-21
<2%	0.3	1.3	2.4	6.6	14.8	1.6
2-2.5%	1.1	2.2	4.3	7.6	15.2	3.0
2.5-3%	2.4	4.0	9.1	13.2	18.4	6.8
3-3.5%	6.6	8.7	14.8	18.1	20.6	13.6
3.5-4%	10.0	16.6	17.7	19.5	13.6	18.8
4-4.5%	17.4	19.3	18.2	17.0	9.9	19.5
4.5-5%	20.9	18.5	17.0	10.0	4.6	15.7
5-5.5%	17.8	13.9	8.5	4.6	1.9	12.1
5.5-6%	12.3	8.9	4.9	2.6	0.8	4.6
6-6.5%	5.9	4.2	1.6	0.1	0.1	2.9
6.5-7%	2.8	1.8	1.2	0.6	0.1	0.8
>7%	2.5	0.6	0.3	0.1	0.0	0.6
Prob[<Mode]	48.3	50.9	50.1	49.2	55.6	50.0

Table 4.5: India : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-SND)

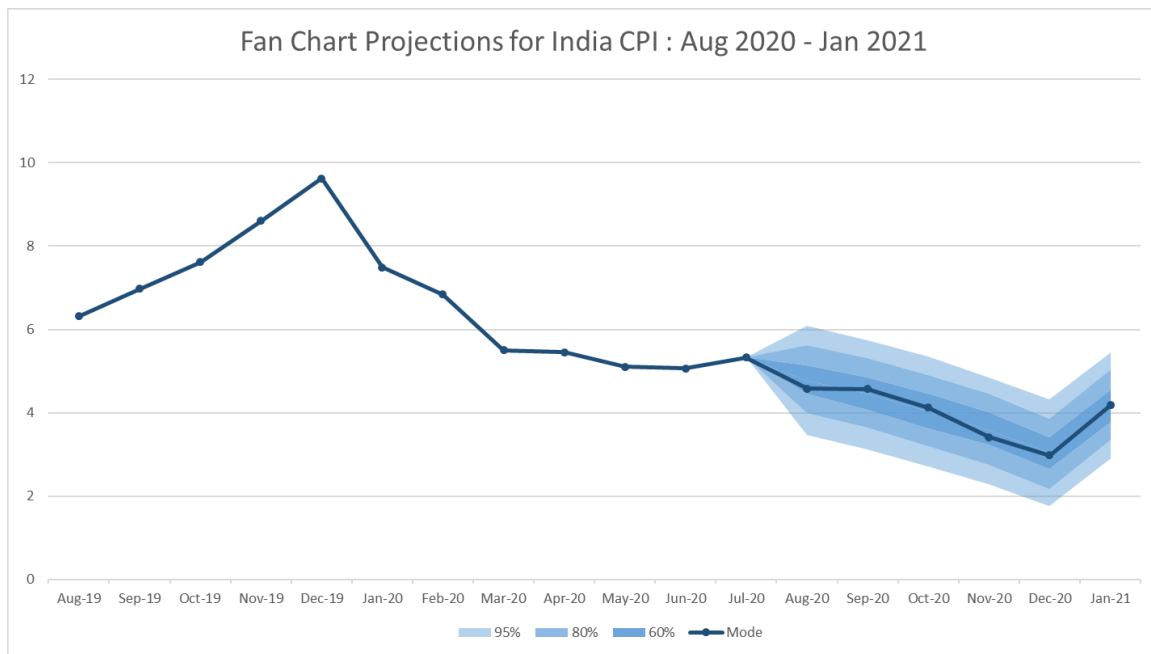


Figure 4.9: India CPI Aug 20 - Jan 21 : Split Normal Distribution

CPI	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20	Jan-20
<1%	0.1	0.0	0.1	0.8	2.2	0.0
1-1.5%	0.1	0.4	0.7	1.3	3.0	0.4
1.5-2%	0.3	0.9	1.6	3.6	7.7	0.7
2-2.5%	0.9	2.2	2.2	6.9	11.2	2.5
2.5-3%	1.2	3.6	7.0	12.9	23.2	5.7
3-3.5%	4.7	7.5	12.5	19.2	21.6	12.2
3.5-4%	9.6	13.3	17.9	22.3	14.1	19.8
4-4.5%	17.8	21.8	22.5	15.5	7.9	24.6
4.5-5%	22.2	20.6	16.7	7.8	5.0	14.2
5-5.5%	17.7	13.4	10.7	5.1	2.4	11.3
5.5-6%	12.5	7.5	4.6	2.2	0.9	5.5
>6%	12.9	8.8	3.5	2.4	0.8	3.1
Actual	8.61	9.63	7.49	6.84	5.5	5.45
Prob[<Mode]	54.0	48.0	44.3	49.7	49.7	53.3

Table 4.6: India : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-GH)

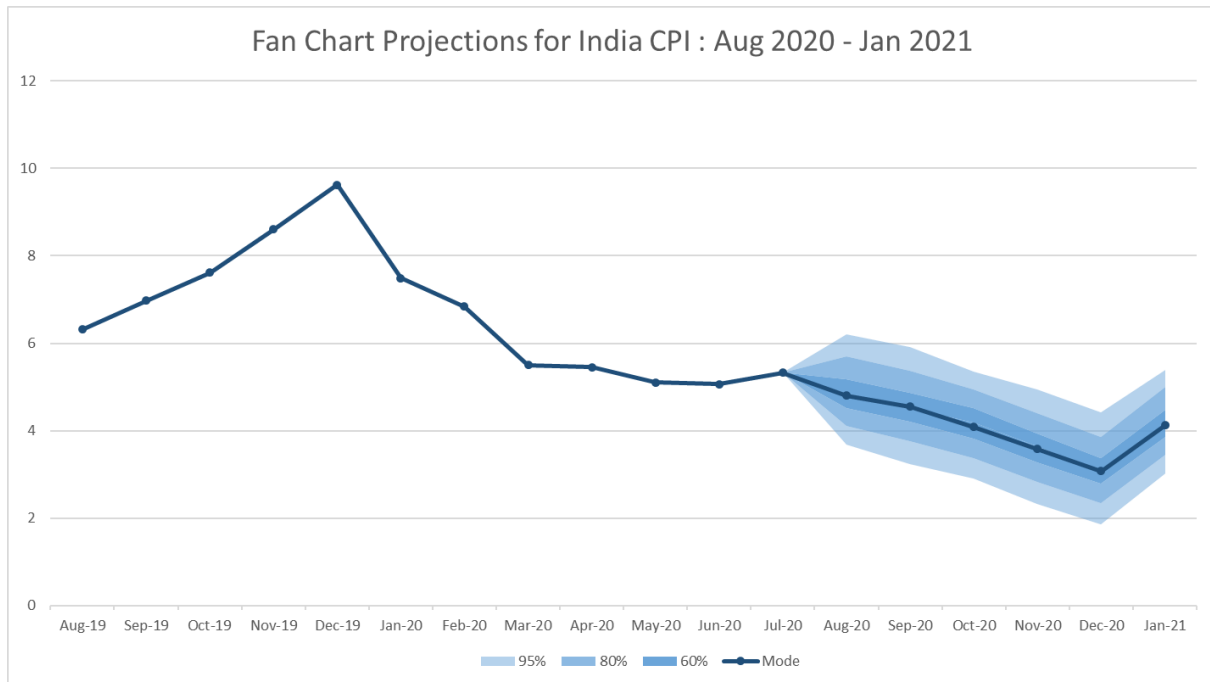


Figure 4.10: India CPI Aug 20 - Jan 21 : Generalised Hyperbolic Distribution

According to Fan Chart based on Split Normal density downward risks to the forecast will remain higher than that of upward risks for all the months except August and November 2020. Fan chart constructed using Generalised Hyperbolic distribution indicates that upward risks would be higher for all the months except August and November 2020. Both of these Fan charts give the most likely interval for the inflation in India for next six months as 3.5% to 5%.

Month	Conf Level	Split Normal Distribution	Generalised Hyperbolic Distribution	Gain in Efficiency
Aug-20	30%	0.6779	0.7004	-0.0225
	60%	1.6324	1.5172	0.1152
	80%	2.6121	2.4787	0.1334
Sep-20	30%	0.7789	0.6567	0.1222
	60%	1.6657	1.5288	0.1369
	80%	2.6203	2.4393	0.1810
Oct-20	30%	0.8392	0.7010	0.1382
	60%	1.7065	1.4889	0.2176
	80%	2.6408	2.4119	0.2289
Nov-20	30%	0.7681	0.7178	0.0503
	60%	1.7082	1.5252	0.1830
	80%	2.5544	2.3995	0.1549
Dec-20	30%	0.7581	0.6694	0.0887
	60%	1.6940	1.5512	0.1428
	80%	2.5647	2.5081	0.0566
Jan-21	30%	0.7855	0.6392	0.1463
	60%	1.6769	1.6078	0.0691
	80%	2.5391	2.5432	-0.0041

Table 4.7: *Comparison of Widths - India: Aug 20 to Jan 21*

The confidence bands using GHD are narrower than that of SND for all except one interval. This shows that GHD is more efficient in terms of reducing the widths of the confidence bands.

4.6.3 Other BRICS Nations

Fan Charts for the next six months are constructed considering the time period Jan 2011 to July 2020. The plot for inflation in all the BRICS nations is given below.

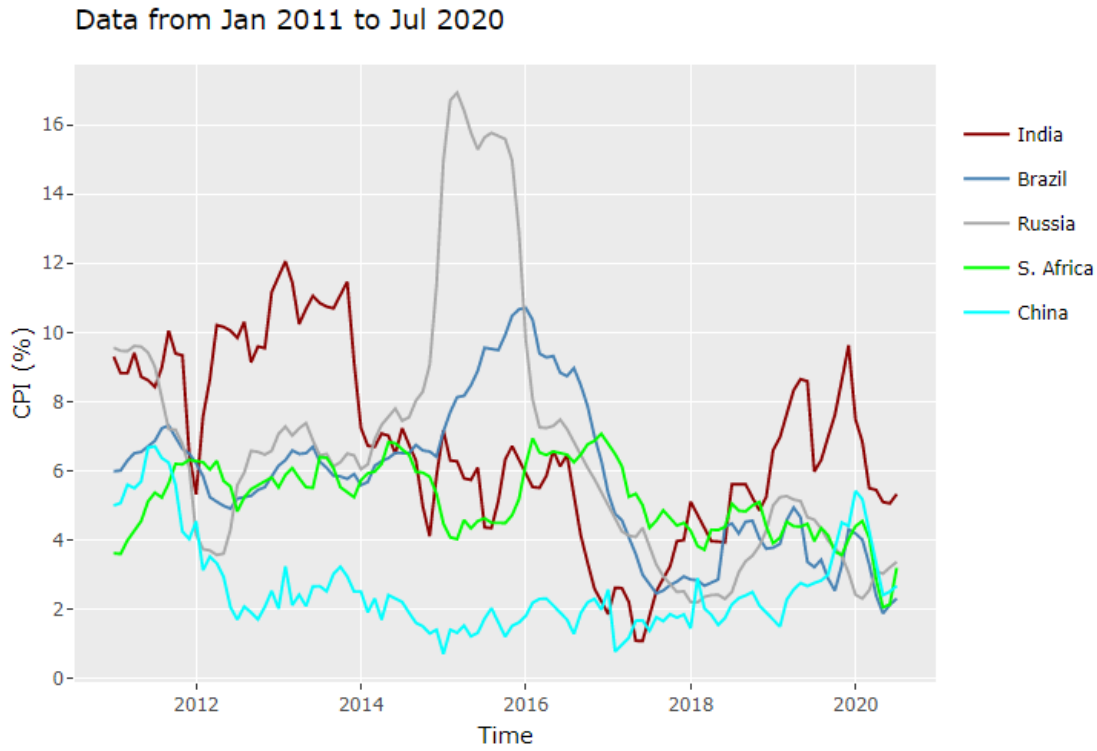


Figure 4.11: *CPI for BRICS Nations*

An R program is developed for computing the probabilities and construction of Fan Charts for the CPI data for each of the countries. The Probability distribution for each case of six-month forecast is given in each case.

Brazil

CPI	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20	Jan-21
<1%	0.0	0.0	0.0	0.0	0.7	0.1
1-1.5%	1.7	0.3	0.0	0.0	8.1	2.6
1.5-2%	17.1	4.0	0.9	2.4	41.3	21.2
2-2.5%	47.4	32.8	16.7	29.1	37.8	46.4
2.5-3%	29.8	45.8	46.1	49.3	11.2	25.3
3-3.5%	3.4	15.9	30.0	17.0	0.9	4.1
3.5-4%	0.6	1.0	5.8	1.9	0.0	0.3
4-4.5%	0.0	0.2	0.5	0.3	0.0	0.0
4.5-5%	0.0	0.0	0.0	0.0	0.0	0.0
>5%	0.0	0.0	0.0	0.0	0.0	0.0
Prob[<Mode]	46.4	48.2	51.7	47.3	44.7	47.6

Table 4.8: *Brazil : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-SND)*

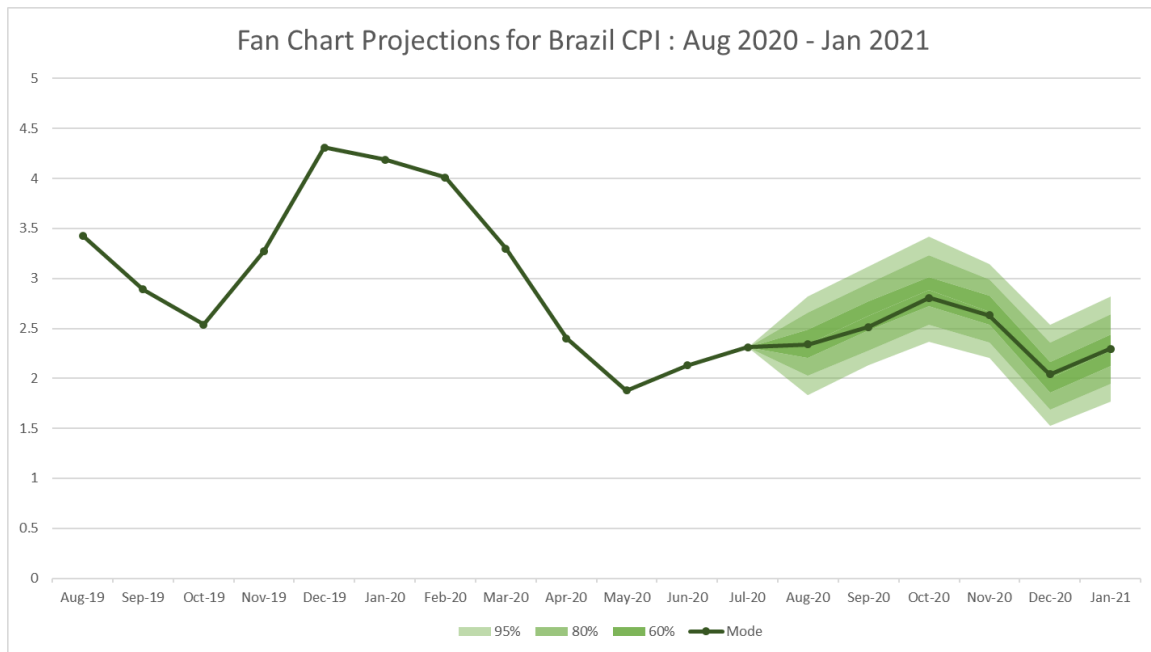


Figure 4.12: *Brazil CPI Aug 20 - Jan 21 : Split Normal Distribution*

CPI	Feb-20	March-20	Apr-20	May-20	Jun-20	Jul-20
<1%	0.0	0.0	0.0	0.0	0.1	0.0
1-1.5%	0.8	0.0	0.0	0.0	4.8	1.4
1.5-2%	14.1	4.1	0.8	1.7	37.2	17.3
2-2.5%	48.0	26.2	11.0	25.8	44.3	49.6
2.5-3%	31.4	48.5	45.2	51.5	12.5	26.6
3-3.5%	4.8	18.5	35.2	18.3	0.8	4.7
3.5-4%	0.8	2.5	7.0	2.3	0.3	0.4
4-4.5%	0.1	0.2	0.7	0.4	0.0	0.0
>4.5%	0.0	0.0	0.1	0.0	0.0	0.0
Actual	8.61	9.63	7.49	6.84	5.5	5.45
Prob[<Mode]	45.3	47.6	44.6	44.8	49.3	35.8

Table 4.9: Brazil : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-GH)

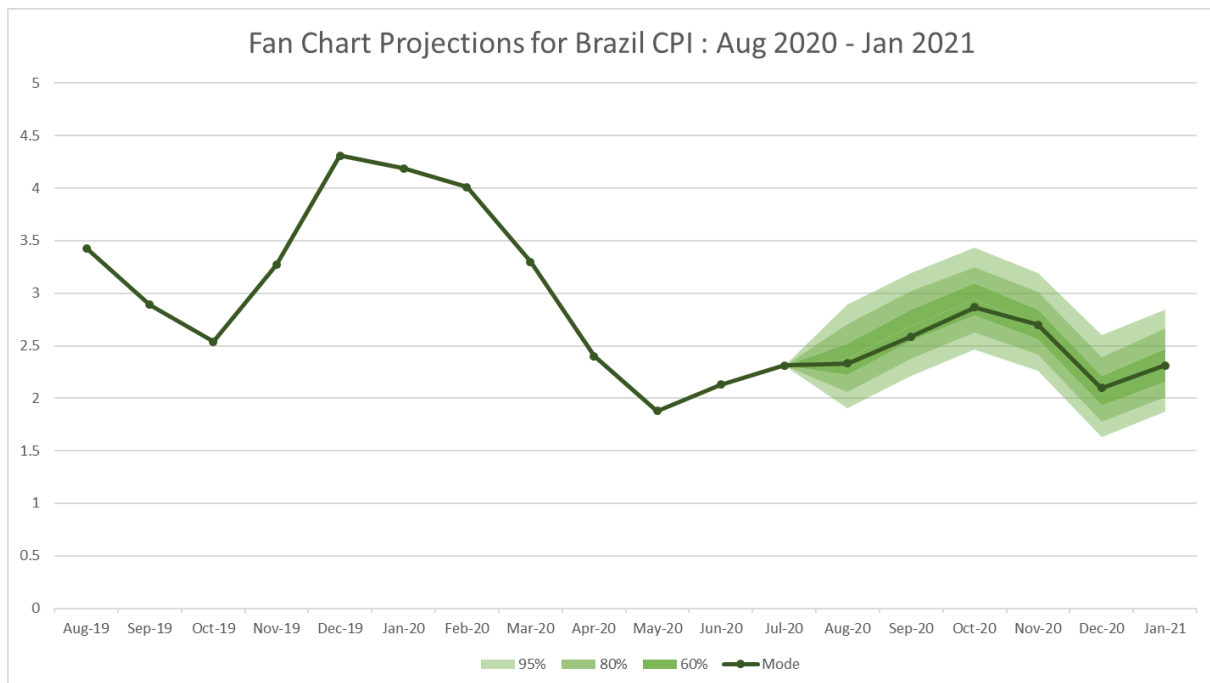


Figure 4.13: Brazil CPI Aug 20 - Jan 21 : Generalised Hyperbolic Distribution

According to Fan Chart based on Split Normal density upward risks to the forecast will remain higher than that of downward risks for all the months except October 2020. Fan chart constructed using Generalised Hyperbolic distribution indicates that upward risks would be higher for all the months. Both of these Fan charts give the most likely interval for the inflation in Brazil for next six months as 3% to 4%.

Month	Conf Level	Split Normal Distribution	Generalised Hyperbolic Distribution	Gain in Efficiency
Aug-20	30%	0.2831	0.3002	-0.0171
	60%	0.6341	0.6488	-0.0147
	80%	0.9884	0.9813	0.0071
Sep-20	30%	0.2944	0.2932	0.0012
	60%	0.6715	0.6455	0.0260
	80%	0.9882	0.9762	0.0120
Oct-20	30%	0.2963	0.3037	-0.0074
	60%	0.6961	0.6219	0.0742
	80%	1.0553	0.9758	0.0795
Nov-20	30%	0.2895	0.2831	0.0064
	60%	0.6300	0.5943	0.0357
	80%	0.9436	0.9307	0.0129
Dec-20	30%	0.3118	0.2881	0.0237
	60%	0.6738	0.6150	0.0588
	80%	1.0093	0.9673	0.0420
Jan-21	30%	0.3163	0.3105	0.0058
	60%	0.6958	0.6503	0.0455
	80%	1.0526	0.9735	0.0791

Table 4.10: *Comparison of Widths - Brazil: Aug 20 to Jan 21*

The confidence bands using GHD are narrower than that of SND for all the intervals except three. This shows that GHD is more efficient in terms of reducing the widths of the confidence bands.

Russia

CPI	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20	Jan-21
<2%	0.9	0.5	0.2	0.0	0.0	0.0
2-2.5%	4.5	1.6	0.9	0.4	0.0	0.1
2.5-3%	15.2	8.1	5.7	3.1	1.0	0.4
3-3.5%	34.5	26.4	18.5	12.3	7.4	1.8
3.5-4%	30.1	35.1	34.1	32.1	23.9	13.8
4-4.5%	10.9	21.8	29.1	33.2	36.2	33.8
4.5-5%	2.9	4.3	7.9	15.1	24.0	33.1
5-5.5%	0.9	1.7	2.4	3.0	5.8	12.2
5.5-6%	0.1	0.5	1.0	0.5	1.6	3.2
6-6.5%	0.0	0.0	0.2	0.3	0.0	1.1
6.5-7%	0.0	0.0	0.0	0.0	0.1	0.5
>7%	0.0	0.0	0.0	0.0	0.0	0.0
Prob[<Mode]	55.6	42.3	59.3	51.2	53.2	43.8

Table 4.11: *Russia : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-SND)*

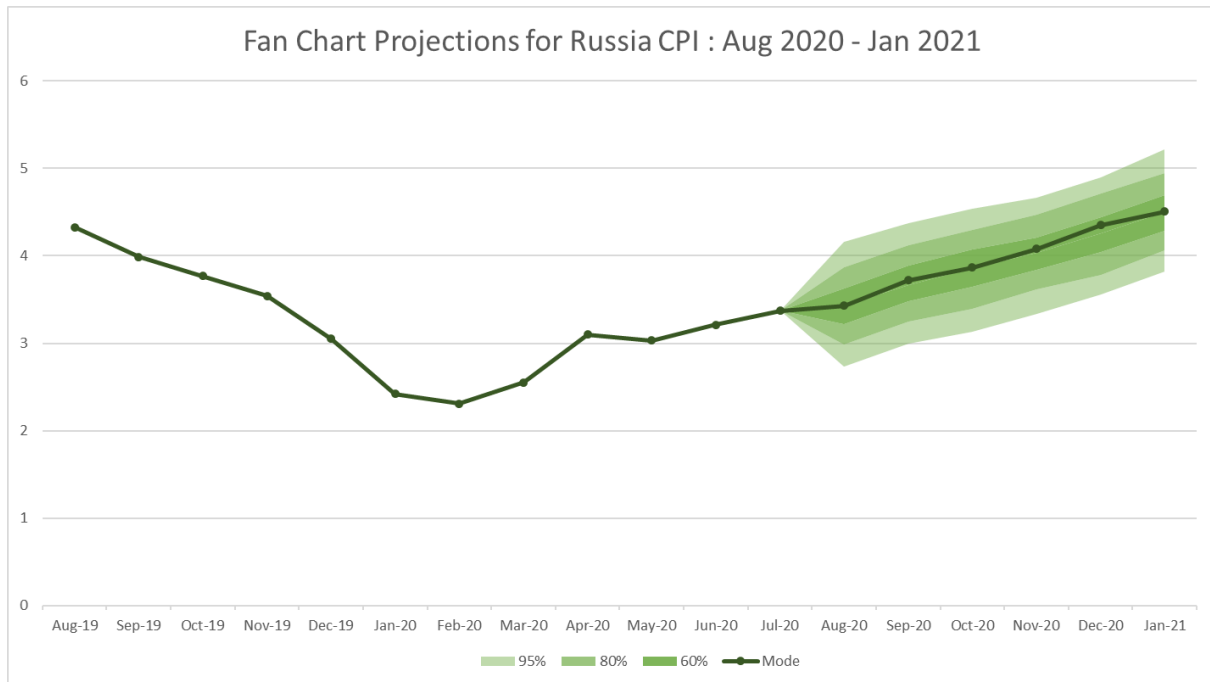


Figure 4.14: *Russia CPI Aug 20 - Jan 21 : Split Normal Distribution*

CPI	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20	Jan-21
<2%	0.3	0.4	0.1	0.0	0.0	0.0
2- 2.5%	1.3	0.5	0.4	0.0	0.1	0.0
2.5-3%	10.3	4.7	2.2	1.6	0.5	0.4
3-3.5%	37.1	18.1	10.0	8.3	3.1	1.6
3.5-4%	33.7	39.9	36.8	29.4	13.8	7.0
4-4.5%	11.7	25.9	34.4	38.5	46.2	28.8
4.5-5%	3.2	7.1	11.4	16.4	25.9	39.4
5-5.5%	2.1	3.0	3.2	4.2	7.0	16.2
5.5-6%	0.3	0.3	1.3	1.1	2.3	3.9
>6%	0.0	0.1	0.2	0.5	1.1	2.7
Prob[<Mode]	40.8	41.0	50.0	45.4	52.5	46.8

Table 4.12: *Russia : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-GH)*

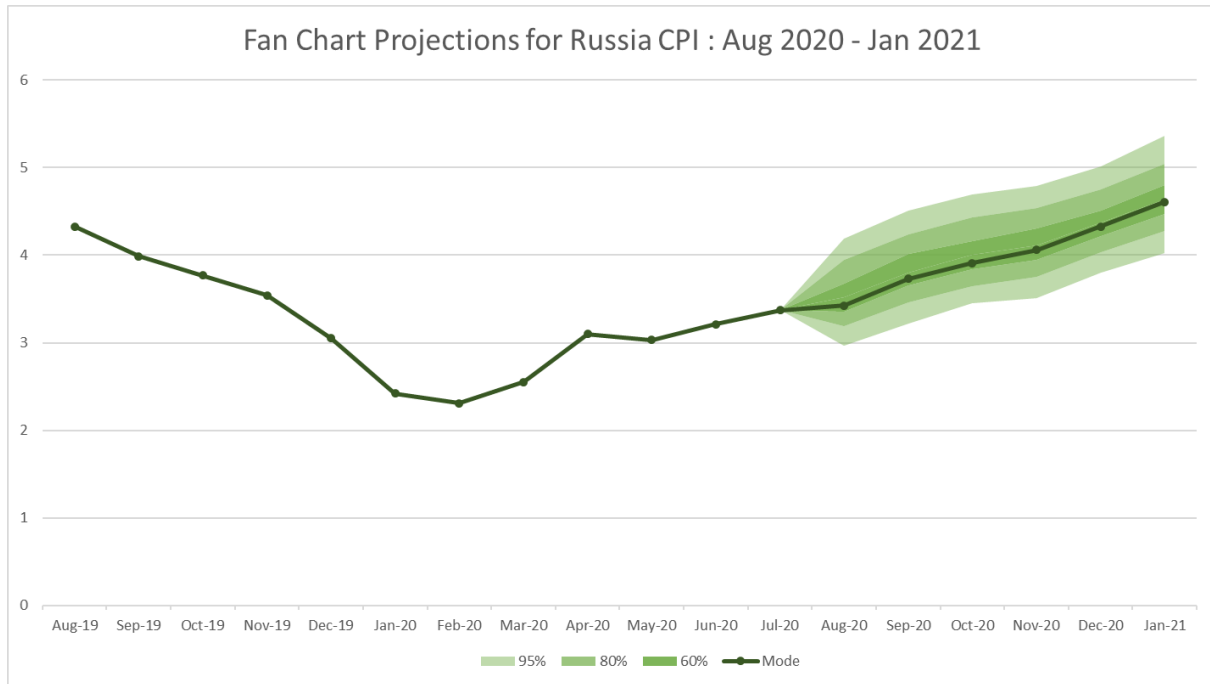


Figure 4.15: *Russia CPI Aug 20 - Jan 21 : Generalised Hyperbolic Distribution*

According to Fan chart based on SND, most likely interval for inflation is 3% to 3.5% for August 2020 and 4% to 4.5% for January 2021. For Fan chart based on GHD, most likely interval for January 2021 is 4.5% to 5%.

Month	Conf Level	Split Normal Distribution	Generalised Hyperbolic Distribution	Gain in Efficiency
Aug-20	30%	0.4016	0.3216	0.0800
	60%	0.8748	0.7541	0.1207
	80%	1.4276	1.2271	0.2005
Sep-20	30%	0.4050	0.3581	0.0469
	60%	0.8756	0.7809	0.0947
	80%	1.3781	1.2949	0.0832
Oct-20	30%	0.4233	0.3216	0.1017
	60%	0.8971	0.7838	0.1133
	80%	1.4085	1.2439	0.1646
Nov-20	30%	0.3700	0.3598	0.0102
	60%	0.8589	0.7884	0.0705
	80%	1.3257	1.2824	0.0433
Dec-20	30%	0.40006	0.2908	0.1092
	60%	0.9354	0.7146	0.2208
	80%	1.3469	1.2131	0.1338
Jan-21	30%	0.4062	0.3304	0.0758
	60%	0.8750	0.7662	0.1088
	80%	1.3912	1.3399	0.0513

Table 4.13: *Comparison of Widths - Russia: Aug 20 to Jan 21*

Here, confidence bands using GHD are narrower than that of SND for all the intervals. This shows that GHD is more efficient in terms of reducing the widths of the confidence bands.

China

CPI	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20	Jan-21
<-1%	0.0	0.0	0.0	0.1	0.0	0.3
-1 - -0.5%	0.0	0.0	0.0	0.4	0.0	3.2
-0.5-0%	0.0	0.0	1.1	4.7	1.5	14.1
0-0.5%	0.0	0.1	4.9	16.2	7.6	32.8
0.5-1%	0.2	0.9	20.6	33.1	23.9	31.2
1-1.5%	1.5	5.8	34.1	30.4	34.9	14.5
1.5-2%	13.0	22.4	27.8	12.4	24.4	3.5
2-2.5%	30.1	36.0	9.3	2.0	6.6	0.4
2.5-3%	33.8	23.5	2.2	0.6	1.0	0.0
3-3.5%	16.2	9.7	0.0	0.1	0.1	0.0
3.5-4%	4.7	1.3	0.0	0.0	0.0	0.0
>4%	0.5	0.3	0.0	0.0	0.0	0.0
Prob[<Mode]	46.3	46.1	51.2	55.7	54.2	41.6

Table 4.14: *China : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-SND)*

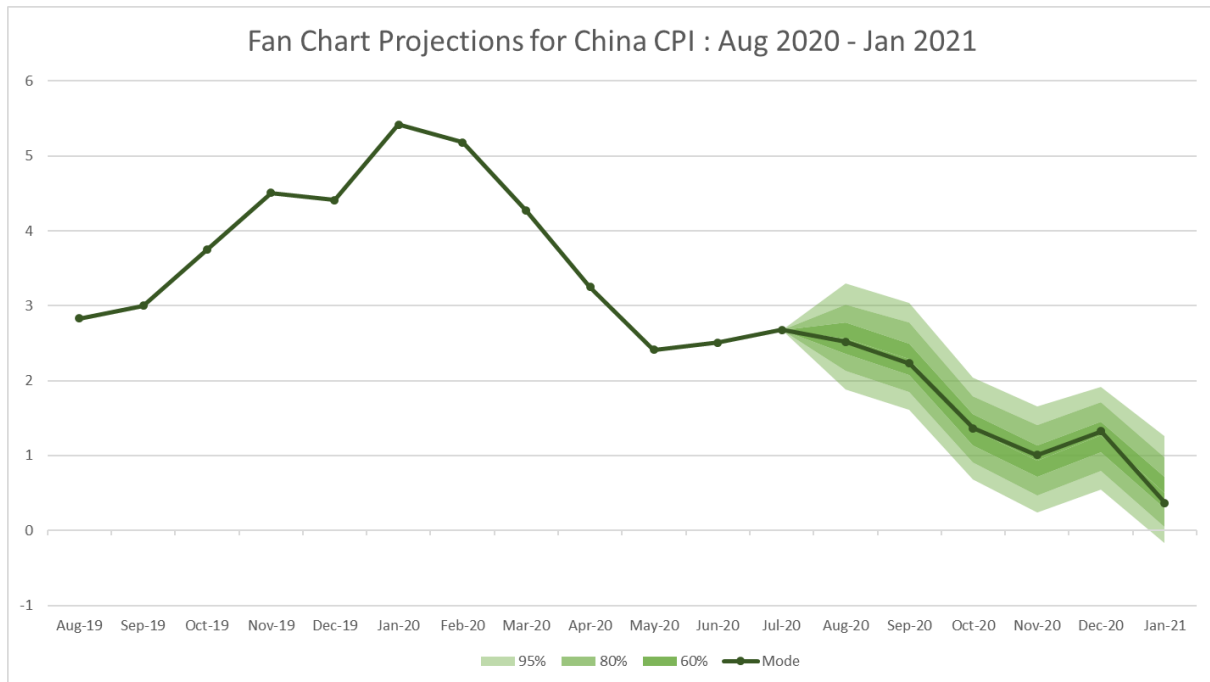


Figure 4.16: *China CPI Aug 20 - Jan 21 : Split Normal Distribution*

CPI	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20	Jan-21
<-1%	0.0	0.0	0.1	0.1	0.1	1.9
-1-(-0.5)%	0.0	0.1	0.3	2.3	1.2	9.3
-0.5-0 %	0.0	0.2	2.9	9.3	4.0	21.6
0-0.5%	0.1	0.8	13.5	27.9	14.2	39.4
0.5-1%	1.5	2.9	30.1	38.3	36.8	22.2
1-1.5%	7.0	11.6	37.4	19.6	32.2	5.1
1.5-2%	20.1	35.6	13.4	2.5	10.0	0.5
2-2.5%	38.4	36.8	1.8	0.0	1.4	0.0
2.5-3%	27.7	9.8	0.5	0.0	0.1	0.0
>3%	5.2	2.2	0.0	0.0	0.0	0.0
Prob[<Mode]	52.3	56.8	60.3	52.8	50.5	56.4

Table 4.15: *China : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-GH)*

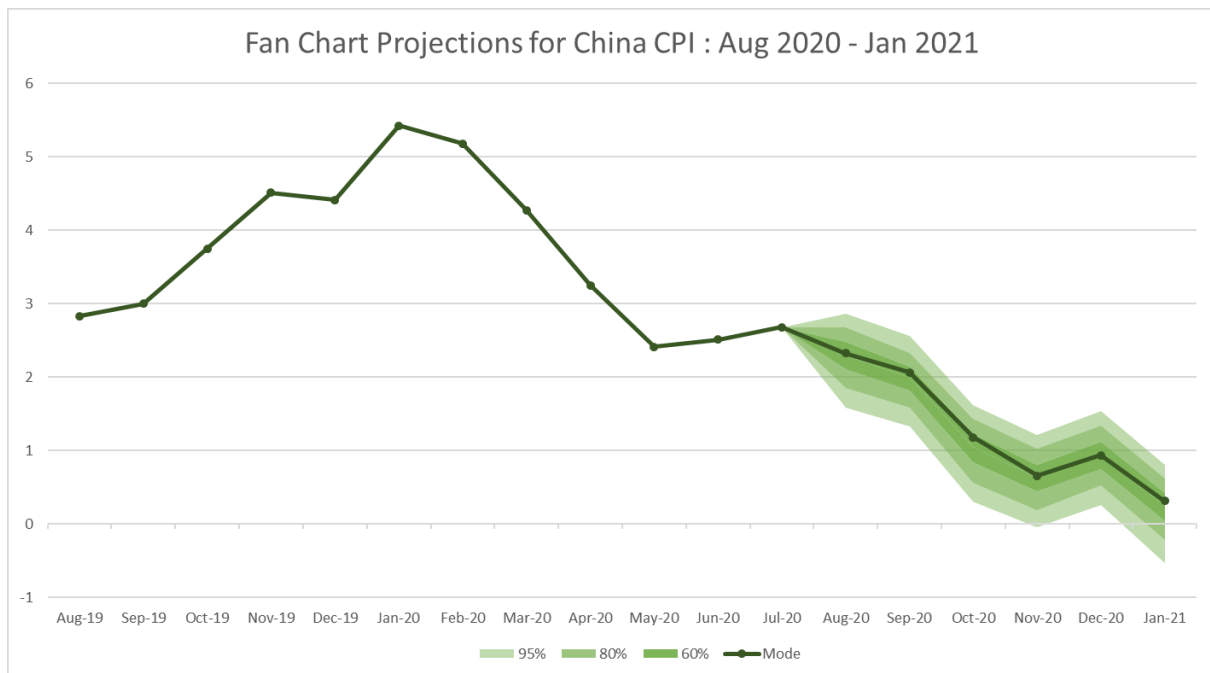


Figure 4.17: *China CPI Aug 20 - Jan 21 : Generalised Hyperbolic Distribution*

According to Fan chart based on SND, China's inflation may hit as low as 0% in January 2021. Almost similar trend is observed in Fan chart constructed using GHD. For October 2020, downward risk is much bigger than the upward risk for the Fan charts using GHD.

Month	Conf Level	Split Normal Distribution	Generalised Hyperbolic Distribution	Gain in Efficiency
Aug-20	30%	0.4124	0.3680	0.0444
	60%	0.8888	0.8307	0.0581
	80%	1.4122	1.2761	0.1361
Sep-20	30%	0.4213	0.3254	0.0959
	60%	0.9367	0.7475	0.1892
	80%	1.4257	1.2373	0.1884
Oct-20	30%	0.4192	0.3914	0.0278
	60%	0.8784	0.8765	0.0019
	80%	1.3672	1.3114	0.0558
Nov-20	30%	0.4230	0.3658	0.0572
	60%	0.9437	0.8438	0.0999
	80%	1.4184	1.2586	0.1598
Dec-20	30%	0.4156	0.3713	0.0443
	60%	0.9142	0.8146	0.0996
	80%	1.3723	1.2724	0.0999
Jan-21	30%	0.4179	0.3771	0.0408
	60%	0.9270	0.8397	0.0873
	80%	1.4240	1.3423	0.0817

Table 4.16: *Comparison of Widths - China: Aug 20 to Jan 21*

Similar to Russia, GHD is outperforming SND for all the intervals. This shows that we gain efficiency if we use GHD instead of SND in the construction of Fan chart.

South Africa

CPI	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20	Jan-21
<1%	0.0	0.0	0.0	0.0	0.0	0.0
1-1.5%	0.0	0.0	0.0	0.0	0.0	0.0
1.5-2%	0.2	0.0	0.0	0.0	0.0	0.0
2-2.5%	2.3	2.2	0.6	0.3	0.4	0.3
2.5-3%	14.8	13.4	3.6	4.3	4.3	3.7
3-3.5%	42.1	37.2	23.2	21.6	27.0	23.6
3.5-4%	33.6	37.3	45.9	41.1	42.5	42.6
4-4.5%	6.5	9.2	21.5	25.8	22.4	24.7
4.5-5%	0.5	0.7	5.0	6.3	3.3	4.9
5-5.5%	0.0	0.0	0.2	0.6	0.1	0.2
5.5-6%	0.0	0.0	0.0	0.0	0.0	0.0
>6%	0.0	0.0	0.0	0.0	0.0	0.0
Prob[<Mode]	53.9	51.6	51.2	38.9	47.0	52.8

Table 4.17: SA : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-SND)

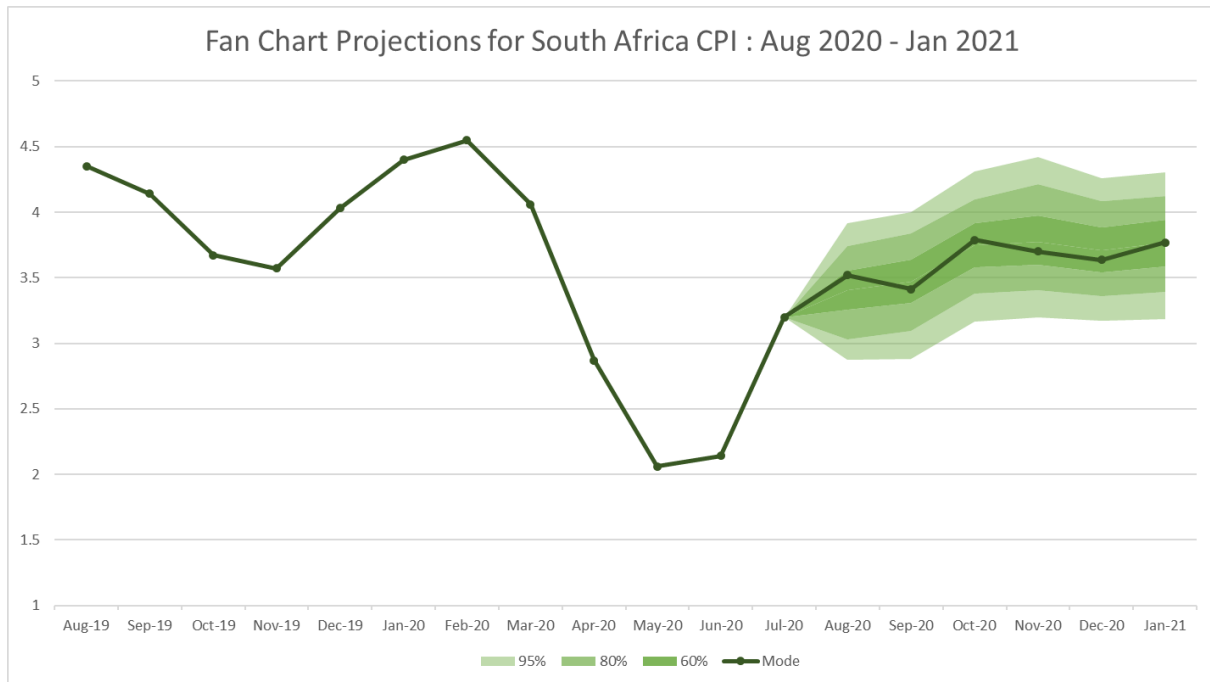


Figure 4.18: S.Africa CPI Aug 20 - Jan 21 : Split Normal Distribution

CPI	Aug-20	Sept-20	Oct-20	Nov-20	Dec-20	Jan-21
$\leq 2\%$	0.3	0.0	0.2	0.0	0.0	0.1
2-2.5%	3.7	2.5	0.9	0.6	1.5	0.4
2.5-3%	18.6	15.4	6.8	5.3	6.8	7.3
3-3.5%	43.7	43.7	27.6	24.9	32.1	26.2
3.5-4%	30.8	32.4	44.2	44.6	42.2	45.2
4-4.5%	2.9	5.8	18.8	22.9	16.2	18.8
$> 4.5\%$	0.0	0.2	1.5	1.7	1.2	2.0
Prob[$<$ Mode]	50.3	46.7	53.3	60.8	56.9	51.1

Table 4.18: SA : Probability Table (in %) - Aug 20 to Jan 21 (SARIMA-GH)

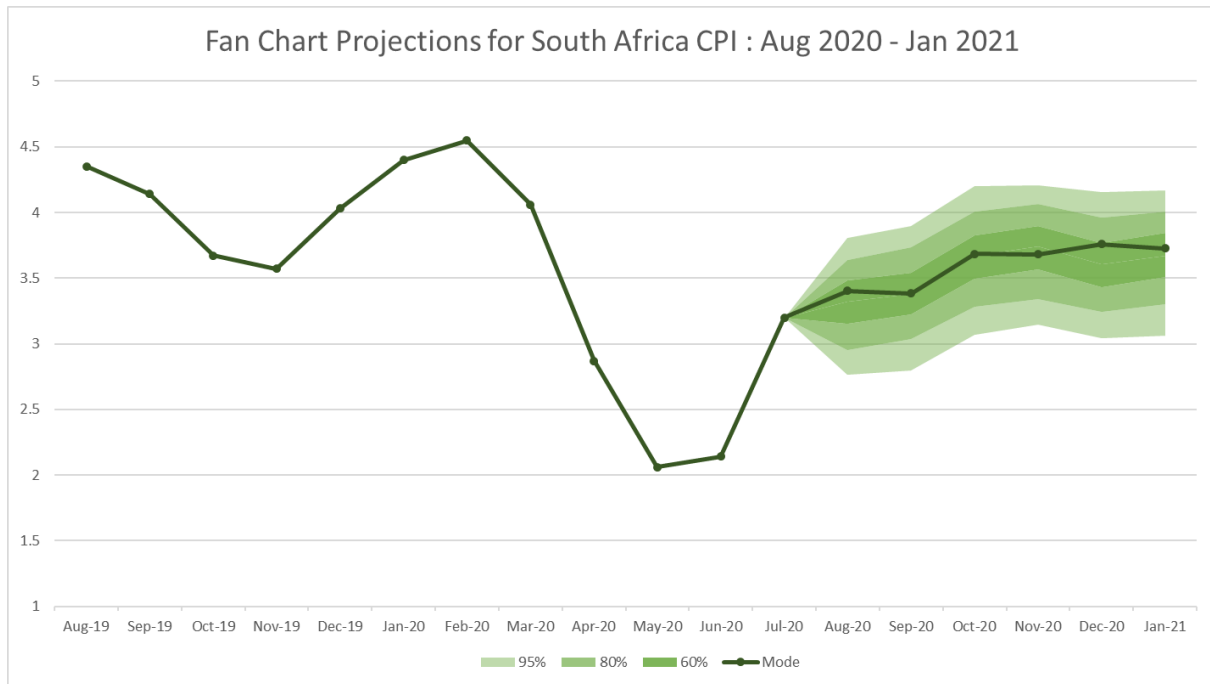


Figure 4.19: S. Africa CPI Aug 20 - Jan 21 : Generalised Hyperbolic Distribution

According to Fan Chart based on Split Normal density upward risks to the forecast will remain higher than that of downward risks in the months of November and December 2020. While, Fan chart constructed using Generalised Hyperbolic distribution indicates that downward risks would be higher for all months except September 2020. Both of these Fan charts give the most likely interval for the inflation in South Africa for next six months as 3% to 4%.

Month	Conf Level	Split Normal Distribution	Generalised Hyperbolic Distribution	Gain in Efficiency
Aug-20	30%	0.3010	0.3291	-0.0281
	60%	0.7060	0.6850	0.0210
	80%	1.0413	1.0409	0.0004
Sep-20	30%	0.3298	0.3160	0.0138
	60%	0.7448	0.7022	0.0426
	80%	1.1212	1.1055	0.0157
Oct-20	30%	0.3318	0.3276	0.0105
	60%	0.7155	0.7231	-0.0076
	80%	1.1492	1.1353	0.0139
Nov-20	30%	0.3768	0.3304	0.0464
	60%	0.8091	0.7226	0.0865
	80%	1.2200	1.0569	0.1631
Dec-20	30%	0.3413	0.3340	0.0073
	60%	0.7307	0.7216	0.0091
	80%	1.0883	1.1189	-0.0306
Jan-21	30%	0.3523	0.3332	0.0191
	60%	0.7273	0.7097	0.0176
	80%	1.1196	1.1036	0.0160

Table 4.19: *Comparison of Widths - South Africa: Aug 20 to Jan 21*

Here, we can see that GHD performs better than that of SND for all the intervals except one. This shows that GHD will give us narrower bands than that of SND, which will increase the efficiency by reducing the uncertainty of the confidence interval.

Chapter 5

Conclusions

5.1 Findings

Fan Chart is a probability distribution for the forecast. The Fan charts helps to communicate the uncertainties clearly to the public, helps in the assessment of risks and to examine its sources for the policy makers. We have used the data on Consumer Price Index (CPI) for the period Jan 2011 to July 2021 for carrying out the inflation forecasting for the next six months.

In addition to the use of Split Normal Distribution as mentioned in the literature, for the inflation forecasting in the Indian context, we have studied some more asymmetrical distributions and have successfully adapted the theory for construction of Fan charts. This is the first attempt for constructing Fan Charts using distributions other than the Split Normal Distribution.

The models SARIMA, VAR and SETAR were fit and compared. The SARIMA model is selected as it has the lowest RMSE. The fan charts were constructed by assuming Split Normal Distribution and Generalised Hyperbolic Distribution. The MLEs for the parameters were estimated. The charts along with the modal class frequencies and confidence interval widths in each case were compared. Both the distributions worked well for this data but Generalised Hyperbolic provided the narrowest confidence intervals, which denotes increased efficiency. Hence for the construction of the Fan chart, it is better to use Generalised Hyperbolic distribution if decision makers want less uncertainty.

This exercise has been carried out for other BRICS nations. This is done to study the effect of COVID-19 pandemic on the Inflation of the developing countries. Brazil's inflation predicted to be remain stable in interval of 2% to 3% with more upward risk to the forecast than the downward risk. For South Africa, the most likely interval is 3% to 4%. For Russia, GHD turns out to be far better choice than the SND as the widths of confidence bands constructed using GHD are smaller for all the confidence

intervals. China's inflation is predicted to hit as low as 0% in January 2021. In all BRICS nations, India's inflation looks to be most uncertain as the confidence intervals have largest widths and also the probabilities of modal classes are low.

5.2 Limitations and Future

It is important to note that the SARIMA model has some limitations. In general, this model fails if there is an unusual growth or downfall at some point in the future due to sudden shocks in the economy. In such cases, some advanced time series models need to be used. Segnon *et al.* (2018) have used STARFIMA-MSM model to forecast the inflation in the G-7 countries. Bayesian VARs (BVARs) with macroeconomic variables were first employed in forecasting by Litterman (1979) and Doan et al. (1984). Since then, VARs and BVARs have been a standard macro-econometric tool routinely used by scholars and policy makers for structural analysis, forecasting and scenario analysis in an ever-growing number of applications. [Agrippino and Ricco (2018)]. These models can be used for modeling the inflation by using different prior distributions. In the era of Artificial Intelligence and Deep Learning, the different models using Neural Networks can be built for the Inflation. Purnima Shaw (2020) and Blix and Sellin (2000) have developed theory related to Bivariate Fan charts. But, their work considered bivariate version of the Split Normal density. Thus, multivariate extension using the other asymmetric distributions remains an area unexplored. Also, there are ample of opportunities regarding the applications of Fan charts in other domains. Especially, considering the current COVID-19 crisis, use of Fan charts in Epidemiology or Public Health related areas would guide the Policy Makers and its stakeholders and can contribute in the betterment of the humanity.

References

1. Banerjee, N. and Das, A. (2011), “Fan Chart: Methodology and its application to inflation forecasting in India”, RBI Working Paper Series, Vol. 5.
2. Barndorff-Nielsen, O. (1977), “Exponentially decreasing distributions for the logarithm of particle size”, Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences, Vol. 353 No. 1674, pp. 401–419.
3. Blix, M. and Sellin, P. (1998), “Uncertainty bands for inflation forecasts”, Sveriges Riksbank Working Paper Series, No. 65.
4. Blix, M. and Sellin, P. (2000), “A bivariate distribution for inflation and output forecasts”, Sveriges Riksbank Working Paper Series, No. 102.
5. Cogley, T., Morozov, S. and Sargent, T.J. (2005), “Bayesian fan charts for U.K. inflation: Forecasting and sources of uncertainty in an evolving monetary system”, Journal of Economic Dynamics and Control, Vol. 29 No. 11, pp. 1893–1925.
6. Dennis, J. and Schnabel, R. (1983), *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, Prentice-Hall, Englewood Cliffs, NJ.
7. John, S. (1982), “The three-parameter two-piece normal family of distributions and its fitting”, Communications in Statistics - Theory and Methods, Vol. 11 No. 8, pp. 879–885.
8. Johnson, N., Kotz, S. and Balakrishnan, N. (1994), *Continuous Univariate Distributions*, John Wiley, New York.
9. Julio, J.M. (2007), “The Fan Chart: The technical details of the new implementation”, Borradores de Economía, Banco de La República, No. 004294.
10. Karsten, P. (1997), “Modelling financial data using generalized hyperbolic distributions”, FDM Preprint, Vol. 48.
11. Karsten, P. (1999), “The generalized hyperbolic model: Estimation, financial derivatives and risk measures”, Verlag Nicht Ermittlbar.

12. Razi, A. and Loke, P.L. (2017), “Fan Chart: The art and science of communicating uncertainty”, IFC Bulletins Chapters, in: Bank for International Settlements (Ed.), Statistical Implications of the New Financial Landscape, Vol. 43.
13. Schnabel, R.B., Koonatz, J.E. and Weiss, B.E. (1985), “A modular system of algorithms for unconstrained minimization”, ACM Transactions on Mathematical Software (TOMS), Vol. 11 No. 4, pp. 419–440.
14. Segnon, M., Bekiros, S. and Wilfling, B. (2018), “Forecasting inflation uncertainty in the G7 countries”, Econometrics, Vol. 6 No. 2, p. 23.
15. Shamsul Alam, A.K.M. and Kamath, S.J. (1986), “Models and forecasts of inflation in a developing economy”, Journal of Economic Studies, Vol. 13 No. 4, pp. 3–29.
16. Shaw, P. (2020), “Construction of bivariate Fan chart from joint distribution”, RBI Working Paper Series , Vol. 02.
17. Shumway, R. and Stoffer, D. (2016), *Time Series Analysis and Its Applications*, Springer International Publishing, New York.
18. Živko, I. and Bošnjak, M. (2018), “Time series modeling of inflation and its volatility in Croatia”, Notitia, No. 3, pp. 1–9.